AI/ML DRIVEN MAINTENANCE SERVICE PROVIDER MATCHING PLATFORM FOR SRI LANKA

INDIVIDUAL THESIS

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Online service marketplaces commonly rely on numerical ratings and aggregated review statistics when recommending service providers; however, these measures provide limited insight into specific service-quality dimensions and may be influenced by suspicious or unreliable customer reviews. This research presents a Bi-LSTM Based Sentiment Analysis and Review Credibility Module that operates as the final review-analysis and re-ranking stage of a service-provider recommendation system. The module receives the Top-10 candidate service providers generated by the preceding recommendation component and analyses their customer reviews using a Multi-Task Bidirectional Long Short-Term Memory (Bi-LSTM) model with a custom Attention Mechanism to classify sentiment across four service-quality aspects: Quality, Punctuality, Communication, and Professionalism. In parallel, an Isolation Forest-based model evaluates review credibility using linguistic and behavioral review characteristics. The aspect-level sentiment outputs are integrated with the corresponding review-credibility information and aggregated at provider level to support trust-aware evaluation and re-ranking. The resulting review-level scores are aggregated at provider level and used to re-rank the initial Top-10 candidates to produce the final Top-5 service-provider recommendations. The module was evaluated using 25,000 customer reviews across fourteen service categories, achieving sentiment-classification accuracies of 96.10%, 87.73%, 86.35%, and 85.58% for Quality, Punctuality, Communication, and Professionalism, respectively, with an average accuracy of 88.94%. The review credibility model achieved 96% overall accuracy and a weighted F1-score of 0.96. The main contribution of this research is the integration of multi-aspect sentiment analysis and review credibility within a unified provider re-ranking process, allowing final provider selection to consider both detailed customer opinions and the estimated reliability of the underlying review evidence.

Keywords: Bi-LSTM, Aspect-Based Sentiment Analysis, Multi-Task Learning, Review Credibility, Service Provider Recommendation

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
ABSA	Aspect-Based Sentiment Analysis
AI	Artificial Intelligence
Bi-LSTM	Bidirectional Long Short-Term Memory
F1	F1-Score
IF	Isolation Forest
LSTM	Long Short-Term Memory
ML	Machine Learning
MHA	Multi-Head Attention
NLP	Natural Language Processing

1 INTRODUCTION

1.1 Overall project context

The increasing use of digital service platforms has changed the way customers identify and select service providers for household and professional service requirements. Such platforms may contain numerous providers offering similar services, making it difficult for customers to determine which providers are relevant and reliable for a particular requirement. Conventional provider-selection approaches commonly depend on profile information, numerical ratings, and basic filtering mechanisms. However, these approaches may not fully capture the relevance of a provider to a user's specific request or the quality of service reflected through previous customer experiences. [1],[5],[10]

The overall research project focuses on developing an intelligent service-provider matching and recommendation system that supports more reliable provider selection by combining multiple sources of information. Rather than relying on a single ranking criterion, the system processes user requirements, provider-related information, historical interaction data, and customer-review evidence through multiple integrated components. The initial recommendation process identifies service providers that are relevant to the user's requirements and produces a set of candidate providers for further evaluation.

Customer reviews provide an additional source of information because they contain detailed opinions regarding previous service experiences. These opinions may describe different dimensions of service performance, including quality, punctuality, communication, and professionalism. However, review information cannot be used effectively by considering only an overall rating or treating every review as equally reliable. Therefore, the overall system incorporates a dedicated review-analysis stage to evaluate both the sentiment expressed within customer reviews and the credibility of the corresponding review evidence. [1],[5],[6]

Within the complete system, the candidate providers generated by the preceding recommendation component are passed to the review-analysis component for further assessment. The review evidence associated with these candidates is analyzed and used

to refine their ranking before the final recommendations are presented to the user. This multi-stage architecture enables the overall project to consider both the relevance of providers to customer requirements and review-derived evidence regarding their previous service performance.

1.2 Individual component context

The individual component developed in this research is the Bi-LSTM Based Sentiment Analysis and Review Credibility Module, which is responsible for analyzing customer-review evidence associated with the candidate service providers identified by the preceding recommendation component. The module operates as a subsequent evaluation stage within the overall service-provider recommendation process. Instead of independently searching the complete provider database, it receives the Top-10 candidate service providers generated by the preceding component and retrieves the customer reviews associated with those candidates for further analysis.

The sentiment-analysis process focuses on four service-quality aspects: Quality, Punctuality, Communication, and Professionalism. A Multi-Task Bidirectional Long Short-Term Memory (Bi-LSTM) model with Attention Mechanism is used to analyze review text and predict sentiment for these four aspects. The multi-task architecture enables the model to learn a shared representation from customer-review text while producing aspect-specific sentiment outputs. The use of bidirectional sequence processing allows contextual information from both directions of a review to be considered, while the attention mechanism supports the identification of relevant information within the learned representation.

In addition to sentiment analysis, the component incorporates a review-credibility assessment process. Customer reviews may contain unusual or unreliable patterns that can distort provider evaluation if every review is given equal influence. Therefore, an Isolation Forest-based model is used to analyze linguistic and behavioral review characteristics and estimate the credibility of the review evidence. This process provides a complementary signal to sentiment analysis by considering the reliability of the information contributing to provider evaluation.

The outputs of the sentiment-analysis and review-credibility processes are subsequently integrated during the scoring stage. Aspect-level sentiment information is combined to obtain an overall sentiment representation for each review, which is then adjusted according to the corresponding credibility information. The resulting review-level scores are aggregated for each candidate provider to obtain provider-level evaluation scores. Based on these scores, the initial Top-10 candidate providers are re-ranked, and the five highest-ranked providers are returned to the application as the final Top-5 service-provider recommendations.

Through this design, the individual component extends provider evaluation beyond overall ratings by considering both aspect-specific customer opinions and the estimated credibility of the underlying review evidence. Its role within the complete system is therefore to provide a review-driven final evaluation of the candidate providers before the final recommendation list is presented to the user.

1.3 Problem statement

Online service-provider recommendation systems commonly use numerical ratings, review counts, provider profiles, and other aggregated indicators to support provider selection. Although these indicators provide a general representation of provider performance, they do not adequately capture the detailed opinions expressed by customers regarding different dimensions of service quality. A provider may perform well in terms of service quality while receiving less favorable feedback regarding punctuality, communication, or professionalism. Representing such experiences through a single overall rating can therefore hide important differences in provider performance that may influence the final selection decision. [1],[5]

Customer reviews provide richer information about previous service experiences; however, directly using review information introduces an additional reliability problem. Reviews may contain unusual, inconsistent, or potentially unreliable patterns, and assigning equal importance to every review may allow such evidence to disproportionately influence provider evaluation. Sentiment analysis alone can determine whether customer opinions are positive, neutral, or negative, but it does not determine how much confidence should be placed in the review from which the

sentiment was obtained. Similarly, review-credibility assessment alone does not identify the specific service-quality dimensions discussed by customers. [6],[7]

Consequently, there is a need for a review-analysis mechanism that can simultaneously represent customer opinions at the aspect level and account for the estimated credibility of the underlying review evidence. In the context of the overall service-provider recommendation system, this requirement becomes particularly important after the initial candidate providers have already been identified. The Top-10 candidate providers may be relevant to the user's request, but their final ordering should also consider the quality and reliability of previous customer experiences.

This research therefore addresses the problem of how to analyze customer reviews across Quality, Punctuality, Communication, and Professionalism, assess the credibility of individual reviews, and integrate these two forms of evidence into a provider-level evaluation mechanism. The Bi-LSTM Based Sentiment Analysis and Review Credibility Module is developed to address this problem by analyzing reviews associated with the initial Top-10 candidate providers and using the resulting sentiment and credibility information to support their re-ranking into the final Top-5 service-provider recommendations.

1.4 Research gap

Existing research has made considerable progress in sentiment analysis, aspect-based sentiment analysis, deep learning-based review classification, and review credibility assessment. Aspect-based sentiment analysis enables customer opinions to be examined at a more detailed level than overall ratings or document-level sentiment by identifying sentiment associated with individual attributes of a service. Deep learning approaches, including Bidirectional Long Short-Term Memory networks and attention mechanisms, further support the extraction of contextual information from review text. Similarly, review credibility and anomaly-detection techniques provide mechanisms for identifying unusual or potentially unreliable review patterns. [1]–[4],[6],[9]

However, these research areas are frequently addressed as separate analytical problems. Aspect-level sentiment models primarily focus on determining what customers express about particular aspects, while review-credibility approaches focus

on assessing whether review evidence exhibits reliable or anomalous characteristics. Applying either process independently leaves an important limitation in the context of service-provider recommendation. Aspect-level sentiment analysis alone may allow sentiment extracted from a low-credibility review to influence provider evaluation in the same manner as sentiment obtained from more credible evidence. Conversely, credibility assessment alone cannot represent customer satisfaction or dissatisfaction across individual dimensions of service performance. [1],[6],[10]

A further gap exists between review-level analysis and the final provider-ranking decision. Producing sentiment predictions or credibility estimates does not by itself determine how these outputs should contribute to the evaluation of a service provider when multiple customer reviews are available. For a recommendation system that has already generated an initial set of relevant candidate providers, a mechanism is required to transform review-level sentiment and credibility information into provider-level evidence that can be used to refine the final ranking.

Therefore, the research gap addressed in this study is the integration of multi-aspect sentiment analysis and review credibility within a unified provider re-ranking process. The proposed Bi-LSTM Based Sentiment Analysis and Review Credibility Module addresses this gap by analyzing customer sentiment across Quality, Punctuality, Communication, and Professionalism, estimating the credibility of the corresponding review evidence, combining these outputs during review-level scoring, and aggregating the resulting information at provider level. This enables the initial Top-10 candidate providers to be re-ranked into a final Top-5 recommendation set using both aspect-specific customer opinions and the estimated credibility of the evidence supporting those opinions.

1.5 Aim and objectives

The aim of this research is to develop a Bi-LSTM Based Sentiment Analysis and Review Credibility Module that analyses customer-review evidence associated with candidate service providers and supports reliable final provider re-ranking by

considering both aspect-level customer sentiment and the estimated credibility of individual reviews.

The specific objectives of this research are:

1. To develop a Multi-Task Bidirectional Long Short-Term Memory (Bi-LSTM) model with an Attention Mechanism for analyzing customer reviews and classifying sentiment across four service-quality aspects: Quality, Punctuality, Communication, and Professionalism.
2. To develop an Isolation Forest-based review credibility model that analyses linguistic and behavioral review characteristics to identify anomalous review patterns and estimate the credibility of individual customer reviews.
3. To integrate aspect-level sentiment predictions with review credibility information so that the influence of customer sentiment on provider evaluation is adjusted according to the estimated credibility of the corresponding review.
4. To aggregate the resulting review-level sentiment and credibility information at provider level and generate an evaluation score for each candidate service provider.
5. To re-rank the Top-10 candidate service providers received from the preceding recommendation component and produce the final Top-5 service-provider recommendations based on the integrated review-analysis results.
6. To evaluate the performance of the developed sentiment analysis and review credibility models using appropriate classification metrics, including Accuracy, Precision, Recall, and F1-score.

1.6 Research questions or hypotheses

Based on the identified research problem and objectives, this study addresses the following research questions:

RQ1: How effectively can a Multi-Task Bidirectional Long Short-Term Memory (Bi-LSTM) model with an Attention Mechanism classify customer-review sentiment across the four service-quality aspects of Quality, Punctuality, Communication, and Professionalism?

RQ2: How effectively can an Isolation Forest-based approach analyze linguistic and behavioral review characteristics to identify anomalous review patterns and estimate the credibility of customer reviews?

RQ3: How can aspect-level sentiment predictions and review credibility information be integrated so that the influence of each customer review on service-provider evaluation reflects both the expressed sentiment and the estimated credibility of the review?

RQ4: How can review-level sentiment and credibility information be aggregated at provider level to support the re-ranking of the Top-10 candidate service providers received from the preceding recommendation component?

RQ5: To what extent can the developed Bi-LSTM Based Sentiment Analysis and Review Credibility Module support the generation of a final Top-5 service-provider recommendation set through the integration of aspect-level sentiment and review credibility information?

1.7 Scope limitations and assumptions

The scope of this research is limited to the Bi-LSTM Based Sentiment Analysis and Review Credibility Module developed as part of the overall service-provider recommendation system. The module operates on the Top-10 candidate service providers received from the preceding recommendation component rather than performing initial provider retrieval from the complete provider database. Customer reviews associated with these candidate providers are analyzed to support the final provider re-ranking process.

The sentiment-analysis scope is limited to four service-quality aspects: Quality, Punctuality, Communication, and Professionalism. A Multi-Task Bi-LSTM model with an Attention Mechanism is used to classify sentiment associated with these aspects. The review credibility process is based on an Isolation Forest model that analyses linguistic and behavioral review characteristics to identify anomalous review patterns. The resulting credibility information is used as an estimate of review reliability and should not be interpreted as definitive evidence that a review is genuine or fraudulent.

The research is evaluated using the developed dataset containing 25,000 customer reviews across fourteen service categories. Consequently, the reported model performance is specific to the dataset, preprocessing procedure, selected service domains, and experimental configuration used in this study. Performance may vary when the models are applied to reviews from different domains, datasets, languages, or distributions. The current research also focuses on textual review evidence and does not attempt to verify reviewer identity or determine the actual intention of a reviewer.

The provider re-ranking process assumes that the preceding recommendation component supplies a valid Top-10 candidate provider set and that sufficient customer-review information is available for those providers. The individual component therefore focuses on analyzing and integrating review-derived evidence rather than evaluating the effectiveness of the initial candidate-generation process. The final Top-5 recommendations are produced by aggregating the sentiment and credibility information obtained from the available reviews associated with each candidate provider.

1.8 Individual contribution

The individual contribution of this research is the design, development, and evaluation of the Bi-LSTM Based Sentiment Analysis and Review Credibility Module within the overall service-provider recommendation system. The component is responsible for analyzing customer-review evidence associated with the Top-10 candidate service providers received from the preceding recommendation component and using this evidence to support the final provider re-ranking process. A Multi-Task Bidirectional Long Short-Term Memory (Bi-LSTM) model with an Attention Mechanism was developed to classify customer sentiment across four service-quality aspects: Quality, Punctuality, Communication, and Professionalism. In addition, an Isolation Forest-based review credibility model was implemented to analyze linguistic and behavioral review characteristics and estimate the credibility of individual reviews. A key contribution of the individual work is the integration of these two analytical processes, where aspect-level sentiment information is combined and adjusted according to review credibility so that reviews with lower estimated credibility have a reduced influence on provider evaluation. The resulting review-level scores are aggregated at

provider level and used to re-rank the initial Top-10 candidate providers into the final Top-5 service-provider recommendations. Therefore, the individual contribution extends beyond independent sentiment classification or review credibility detection by integrating both forms of review evidence into a unified provider-level evaluation and re-ranking process.

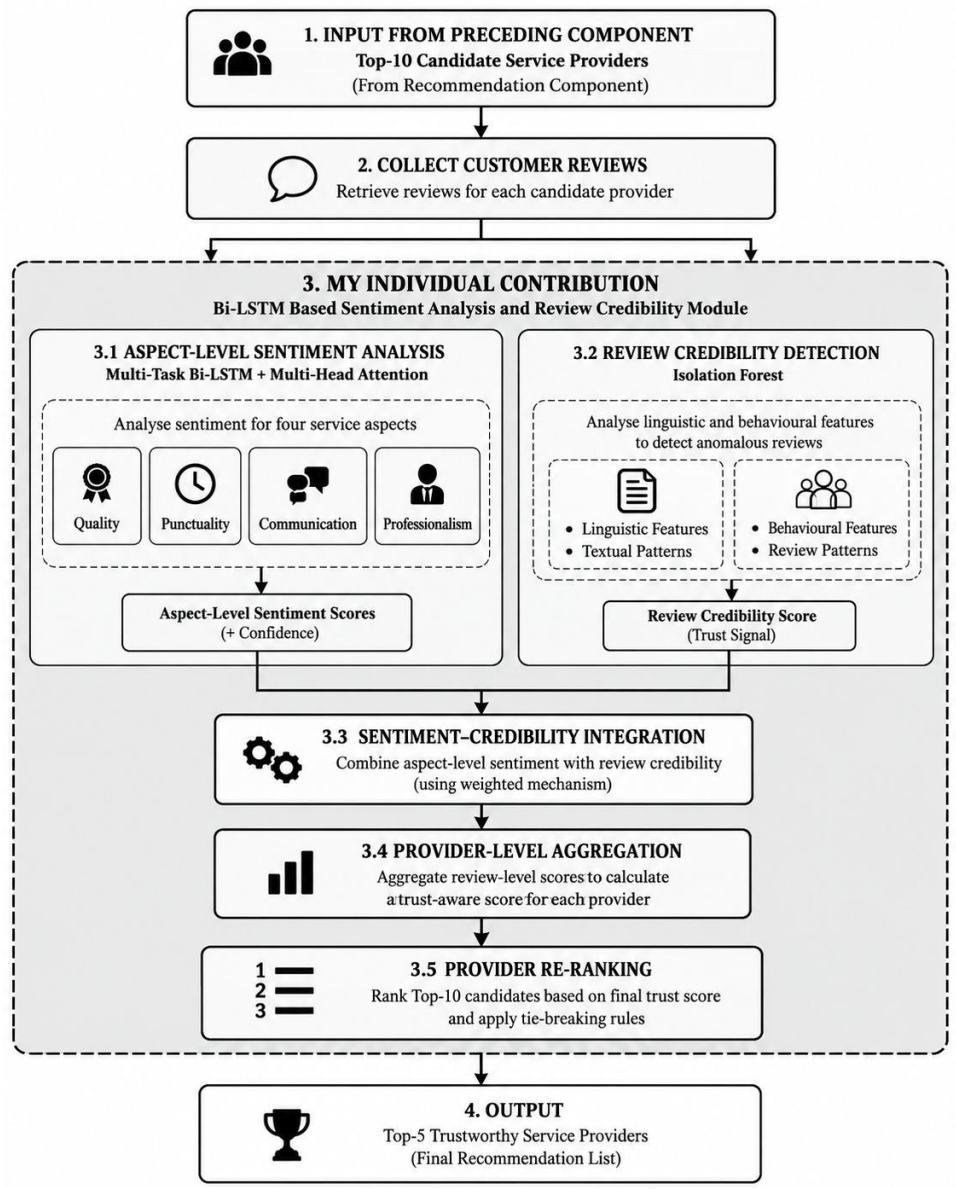


Figure 1.1:Overview of the individual contribution of the Bi-LSTM Based Sentiment Analysis and Review Credibility Module.

1.9 Thesis structure

This thesis is organized into eight chapters. Chapter 1 introduces the overall research project and the individual Bi-LSTM Based Sentiment Analysis and Review Credibility Module, including the problem statement, research gap, aim and objectives, research questions, scope and limitations, and individual contribution. Chapter 2 presents the literature review and related work relevant to aspect-based sentiment analysis, Bi-LSTM architectures, attention mechanisms, multi-task learning, review credibility assessment, anomaly detection, and service-provider recommendation, and establishes the research gap addressed by this study. Chapter 3 describes the research methodology, including the dataset, data preparation and preprocessing, Multi-Task Bi-LSTM with Attention Mechanism model, Isolation Forest-based review credibility approach, sentiment and credibility integration, provider-level score aggregation, and evaluation methods. Chapter 4 presents the product and solution design, including the requirements, component architecture, system interactions, and detailed design of the individual component within the overall service-provider recommendation system. Chapter 5 describes the implementation, integration, and testing of the component, including the development environment, model implementation, Firebase-based data integration, interaction with the preceding recommendation component, and verification of the Top-10 to Top-5 provider re-ranking process. Chapter 6 presents the experimental and technical results obtained from the sentiment analysis and review credibility models and the integrated component. Chapter 7 discusses the findings in relation to the research objectives and existing literature, identifies the individual research and product contributions, and discusses the limitations and implications of the study. Finally, Chapter 8 concludes the thesis by summarizing the major findings, evaluating the achievement of the research objectives, and presenting potential directions for future improvements.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This literature review examines research relevant to the development of the Bi-LSTM Based Sentiment Analysis and Review Credibility Module, with particular emphasis on the analysis of customer reviews for service-provider evaluation. The review focuses on two closely related research areas: aspect-level sentiment analysis and review credibility assessment. These areas are examined together because the proposed component requires not only an understanding of customer opinions regarding specific dimensions of service quality, but also an assessment of the reliability of the review evidence contributing to provider evaluation.

The scope of the review covers Natural Language Processing (NLP) and machine-learning approaches used for sentiment classification, with particular attention to Aspect-Based Sentiment Analysis (ABSA), recurrent neural networks, Long Short-Term Memory (LSTM) networks, Bidirectional Long Short-Term Memory (Bi-LSTM) architectures, attention mechanisms, and multi-task learning approaches [1]–[5]. These techniques are reviewed to establish the theoretical and technical basis for analyzing customer sentiment across the four service-quality aspects considered in this research: Quality, Punctuality, Communication, and Professionalism.

The review also examines existing approaches to review credibility assessment, suspicious-review identification, anomaly detection, and the use of linguistic and behavioral characteristics for evaluating review reliability [6]–[8]. Particular consideration is given to unsupervised anomaly-detection techniques, including Isolation Forest, due to their applicability when reliable ground-truth labels for suspicious reviews are limited or unavailable[9]. Studies that combine review-derived information with recommendation and ranking mechanisms are further considered to determine how sentiment and credibility evidence have previously been incorporated into decision-support systems[10],[11].

Relevant literature is analyzed comparatively rather than only descriptively. Existing approaches are examined according to their research objectives, datasets, review representations, modelling techniques, evaluation methods, reported performance, and

identified limitations. Particular attention is given to whether existing studies perform overall or aspect-level sentiment analysis, whether multiple service aspects are modelled simultaneously, whether review credibility is considered, and whether the resulting information contributes directly to provider-level ranking or recommendation.

The review is therefore structured to establish the theoretical foundations of the individual component, critically examine related research and existing techniques, and identify limitations in current approaches. These findings are subsequently synthesized to establish the research gap addressed by integrating multi-aspect sentiment analysis and review credibility within the provider re-ranking process developed in this study.

2.2 Domain and theoretical foundations

2.2.1 Aspect-Based Sentiment Analysis

Sentiment analysis is a Natural Language Processing (NLP) task concerned with identifying and interpreting opinions, attitudes, and emotional orientation expressed in textual content. Traditional sentiment-analysis approaches commonly assign an overall sentiment label to a complete sentence, review, or document. Although document-level sentiment provides a general indication of customer satisfaction, it may fail to represent situations in which a customer expresses different opinions regarding different characteristics of the same service. Aspect-Based Sentiment Analysis (ABSA) addresses this limitation by analyzing sentiment with respect to individual attributes or aspects discussed within a review [1],[5].

Aspect-level analysis is particularly relevant to service-provider evaluation because customer experience is multidimensional. A customer may provide a positive opinion regarding the quality of completed work while simultaneously expressing dissatisfaction with punctuality or communication. Representing the entire review using a single sentiment value may therefore conceal information that is important for evaluating provider performance. In this research, customer-review sentiment is analyzed across four service-quality aspects: Quality, Punctuality, Communication, and Professionalism. This enables the review-analysis component to preserve

differences among important dimensions of service performance before the resulting sentiment evidence is used for provider evaluation.

2.2.2 Bidirectional Long Short-Term Memory Networks

Long Short-Term Memory (LSTM) networks are a form of recurrent neural network designed to model sequential information while reducing difficulties associated with learning long-term dependencies in conventional recurrent architectures [1],[3]. Their internal memory and gating mechanisms allow relevant information to be retained or discarded as a sequence is processed, making LSTM-based architectures suitable for textual data in which the interpretation of a word can depend on surrounding context.

A Bidirectional Long Short-Term Memory (Bi-LSTM) network extends this capability by processing an input sequence in both forward and backward directions [1],[3]. The forward network captures information based on preceding tokens, while the backward network incorporates information from subsequent tokens. Their representations can then be combined to provide a context-sensitive representation of the input text. This characteristic is useful for customer-review analysis because the sentiment associated with a service aspect may depend on words occurring both before and after the relevant expression. Bi-LSTM therefore provides the sequential modelling foundation used for sentiment analysis in the proposed individual component.

2.2.3 Attention Mechanism

Although recurrent architectures can represent sequential dependencies, not every part of a customer review contributes equally to a sentiment decision. Attention mechanisms address this issue by allowing a model to assign different levels of importance to different elements of an input representation when producing an output [3],[4]. This enables the model to focus more strongly on information that is relevant to the prediction task.

In recurrent sentiment models, an attention mechanism can learn a weighted representation of sequential outputs so that information contributing more strongly to a prediction receives greater emphasis [3],[4]. In this research, a custom Attention Mechanism is applied to the sequential Bi-LSTM representations before the shared dense layer. The mechanism complements bidirectional sequence modelling by

producing a weighted review representation that is subsequently used by the four aspect-specific sentiment outputs.

2.2.4 Multi-Task Learning

Multi-Task Learning (MTL) is a machine-learning paradigm in which multiple related prediction tasks are learned jointly rather than through completely independent models [2]. A shared representation can capture information that is useful across related tasks, while task-specific output layers learn the characteristics required for individual predictions. Knowledge sharing among related tasks may improve representation learning and reduce unnecessary duplication of separate models.

The four aspect-level sentiment tasks considered in this research are related because they are derived from the same customer-review text and describe different dimensions of a common service experience. The proposed sentiment model therefore uses a multi-task architecture in which textual representations are shared while aspect-specific sentiment predictions are produced for Quality, Punctuality, Communication, and Professionalism. This provides the theoretical basis for analyzing several service aspects within a unified sentiment-analysis model instead of treating each aspect as an entirely isolated review-analysis problem.

2.2.5 Review Credibility Assessment

The usefulness of customer reviews for recommendation depends not only on the sentiment they express but also on the reliability of the evidence they provide. Online review environments may contain reviews exhibiting unusual linguistic or behavioral characteristics. If every review contributes equally to provider evaluation, anomalous or unreliable review evidence may influence the resulting provider score regardless of its credibility.

Review credibility assessment therefore aims to evaluate characteristics that can provide an indication of the reliability of review evidence [6],[7]. Such assessment can consider textual or linguistic characteristics of a review together with behavioral information associated with review activity, depending on the data available. Importantly, an estimated credibility score should not automatically be interpreted as

definitive proof that a review is genuine or fraudulent. Rather, it represents an analytical signal that can be incorporated into subsequent decision-making.

Within the proposed component, credibility analysis complements aspect-level sentiment analysis. Sentiment analysis identifies the nature of customer opinions, whereas credibility assessment provides information regarding the degree of confidence that should be assigned to the review evidence contributing those opinions. This relationship provides the foundation for integrating sentiment and credibility during provider-level evaluation.

2.2.6 Isolation Forest for Review Credibility Analysis

Isolation Forest is an unsupervised anomaly-detection technique based on the principle that anomalous observations are generally easier to isolate than normal observations [9]. Instead of directly modelling normal behavior, the method constructs multiple random isolation trees by recursively selecting features and split values. Observations requiring relatively fewer partitions to become isolated can be considered more anomalous than observations embedded within dense regions of the data.

This property makes Isolation Forest applicable to review credibility assessment when review instances can be represented through relevant linguistic and behavioral characteristics. Reviews exhibiting patterns that differ substantially from the majority of observed review behavior can be assigned stronger anomaly indications. The resulting information can subsequently contribute to the estimation of review credibility.

In this research, the Isolation Forest-based credibility model operates alongside the Multi-Task Bi-LSTM sentiment model rather than replacing sentiment analysis. The two models address different characteristics of customer-review evidence: the sentiment model determines what is expressed regarding the four service aspects, while the credibility model evaluates whether the corresponding review exhibits potentially anomalous characteristics. Their outputs are subsequently integrated within the individual component to support provider evaluation and re-ranking.

2.3 Related research

Previous research related to customer-review intelligence has primarily developed along two research directions: sentiment-based opinion analysis and review credibility or suspicious-review detection. These studies demonstrate the potential of machine learning and deep learning techniques for extracting useful information from online reviews. However, the objectives and outputs of the two research directions are generally different, creating an opportunity to investigate their combined use in service-provider evaluation.

2.3.1 Deep Learning for Sentiment Analysis

Deep learning approaches have been widely investigated for sentiment classification because of their ability to learn representations directly from textual sequences. Recurrent neural architectures, particularly Long Short-Term Memory (LSTM) and Bidirectional Long Short-Term Memory (Bi-LSTM) networks, have been applied to review classification tasks to capture contextual relationships between words [1],[3],[4]. Compared with conventional feature-based classifiers, these approaches can reduce dependence on manually designed textual features and provide stronger representations of sequential review content.

Bi-LSTM-based approaches are particularly relevant when the interpretation of an opinion depends on contextual information appearing before and after a particular expression. Previous studies have therefore combined bidirectional recurrent architectures with word representations and other neural mechanisms for sentiment classification [3],[4]. Nevertheless, many existing approaches primarily focus on predicting an overall sentiment label for a complete review. Such document-level classification provides limited information when the same review contains different opinions regarding multiple characteristics of a service.

2.3.2 Aspect-Based Sentiment Analysis

Aspect-Based Sentiment Analysis (ABSA) has been investigated to overcome the limitations of assigning a single sentiment label to an entire review. Existing studies have explored machine-learning and deep-learning approaches for identifying sentiment associated with specific product or service attributes [1],[5]. This enables customer feedback to be represented at a finer level of granularity and is particularly

useful in domains where an overall customer experience consists of several distinct dimensions.

Attention-based architectures have also been investigated in ABSA because sentiment-relevant information may occur at different locations within a review [3],[4]. Attention mechanisms allow models to emphasize relevant textual representations during classification rather than treating all parts of a sequence as equally informative. More recent approaches have combined recurrent networks with attention mechanisms to improve contextual representation and aspect-oriented sentiment prediction[3],[4].

Although these studies demonstrate the effectiveness of aspect-level analysis, many primarily evaluate the accuracy of sentiment classification itself. The resulting aspect predictions are not always incorporated into a subsequent provider-level ranking mechanism. Furthermore, sentiment evidence may be processed without considering whether the review supplying that evidence exhibits potentially unreliable characteristics.

2.3.3 Multi-Task Sentiment Classification

Multi-task learning has also been investigated for related NLP classification problems in which several prediction tasks share common textual information [2]. Instead of training completely independent models, a multi-task architecture can learn shared representations and use task-specific outputs for individual predictions. This approach is relevant to multi-aspect review analysis because different service aspects may occur within the same textual review and share contextual information.

For service-provider reviews, Quality, Punctuality, Communication, and Professionalism represent distinct but related dimensions of the customer experience. A shared sentiment-analysis architecture can therefore exploit common review representations while maintaining separate predictions for individual aspects. However, the use of multi-task aspect sentiment predictions alone still does not address the reliability of the review evidence from which those predictions are derived.

2.3.4 Review Credibility and Suspicious-Review Detection

A separate body of research has investigated fake, deceptive, spam, suspicious, or low-credibility online reviews. Existing studies have examined textual characteristics,

reviewer behavior, rating patterns, review frequency, and other metadata to distinguish normal review behavior from potentially suspicious activity [6],[8]. Both supervised classification and unsupervised anomaly-detection techniques have been explored for this purpose.

Supervised approaches can achieve effective classification when sufficiently reliable labelled datasets are available; however, obtaining trustworthy labels for deceptive or suspicious reviews can be challenging [6],[7]. This limitation has motivated the investigation of unsupervised approaches that identify observations differing from dominant patterns without requiring extensive class-labelled data.

Isolation Forest has been used in anomaly-detection applications because it isolates unusual observations through random partitioning rather than requiring an explicit model of normal behavior [9]. This property provides a useful foundation for analyzing reviews represented through linguistic and behavioral characteristics. Nevertheless, detecting an anomalous review and analyzing the opinion expressed by that review remain separate problems. Anomaly detection alone cannot determine whether a customer expresses positive, neutral, or negative sentiment regarding a particular service-quality aspect.

2.3.5 Sentiment and Credibility in Recommendation

Research on recommender systems has increasingly considered review text as an additional information source beyond numerical ratings [10],[11]. Sentiment extracted from customer reviews can provide evidence regarding user experiences and item or service characteristics, while credibility-related information can potentially reduce the influence of unreliable evidence. These findings indicate that both forms of information are relevant to recommendation quality.

However, a limitation remains when sentiment analysis and credibility assessment are implemented as independent processes. If aspect sentiment is incorporated into ranking without considering credibility, potentially unreliable review evidence may receive the same influence as more credible evidence. Conversely, credibility assessment without aspect-level sentiment does not describe how customers evaluate individual dimensions of provider performance. In addition, many studies concentrate

on review classification or credibility detection as the final analytical objective rather than transforming their combined outputs into provider-level evidence for re-ranking an existing candidate set.

Therefore, the present research builds upon these related areas by combining a Multi-Task Bi-LSTM with Attention Mechanism for sentiment analysis across Quality, Punctuality, Communication, and Professionalism with an Isolation Forest-based review credibility process. Rather than treating the outputs as independent predictions, the Bi-LSTM Based Sentiment Analysis and Review Credibility Module integrates the resulting evidence, aggregates it at provider level, and uses it to re-rank the Top-10 candidate providers into the final Top-5 recommendations.

2.4 Existing systems products and techniques

Existing online service and recommendation platforms commonly use numerical ratings, review counts, provider profiles, popularity indicators, and user preferences to support provider or item selection. Although these mechanisms provide a convenient representation of overall customer satisfaction, numerical ratings alone provide limited information regarding the individual dimensions of a service experience. Two providers with similar overall ratings may differ considerably in terms of service quality, punctuality, communication, or professionalism. Consequently, review-text analysis provides an opportunity to obtain more detailed evidence that cannot be represented adequately through aggregated ratings alone [1],[5].

Existing sentiment-analysis techniques range from traditional machine-learning approaches to deep neural architectures. Conventional approaches generally rely on manually engineered textual features combined with classifiers such as Support Vector Machines, Naive Bayes, or Logistic Regression [1],[3],[4]. While these methods can provide effective baseline performance, their ability to represent complex contextual relationships within customer reviews is limited by the selected feature representation. Deep-learning approaches, particularly LSTM and Bi-LSTM architectures, address this limitation by learning sequential representations directly from review text. Attention mechanisms further enhance these architectures by enabling greater

emphasis to be placed on information that is relevant to a particular prediction [1],[3],[4].

For aspect-oriented review analysis, existing techniques attempt to distinguish opinions associated with individual attributes instead of assigning only a single sentiment label to an entire review. This is particularly relevant to service-provider evaluation because a customer can express different sentiments regarding multiple dimensions within the same service experience. Multi-task learning provides an additional mechanism for jointly modelling related aspect-level classification tasks through shared representations while maintaining task-specific predictions [2],[5]. However, these approaches primarily address what customers express and do not inherently evaluate the credibility of the reviews from which the sentiment information is obtained.

Review credibility and suspicious-review detection techniques address a different limitation of online review systems. Existing approaches use combinations of linguistic features, review patterns, behavioral characteristics, rating behavior, and other available metadata to identify potentially unusual review activity [6],[9]. Supervised classification techniques require labelled examples of credible and suspicious reviews, whereas unsupervised anomaly-detection methods can identify observations that deviate from dominant patterns without requiring extensive labelled datasets. Isolation Forest represents one such anomaly-detection technique and can be applied to review representations to identify unusual linguistic or behavioral patterns[6],[9]. Nevertheless, an anomaly indication should be considered an estimate of credibility rather than definitive evidence that a review is fraudulent.

The principal limitation observed across these existing techniques is that sentiment analysis and review credibility assessment frequently address separate analytical objectives. Sentiment-analysis approaches can provide detailed information about customer opinions but may treat review evidence without considering its estimated reliability. Credibility-analysis approaches can identify anomalous review characteristics but do not determine customer sentiment regarding individual service dimensions. Furthermore, producing sentiment labels or credibility estimates does not

automatically translate these outputs into provider-level evidence suitable for re-ranking candidate service providers.

The Bi-LSTM Based Sentiment Analysis and Review Credibility Module developed in this research combines these complementary techniques within a single review-analysis process. Multi-Task Bi-LSTM with Attention Mechanism is used for sentiment classification across Quality, Punctuality, Communication, and Professionalism, while an Isolation Forest-based process provides review credibility information. These outputs are subsequently integrated and aggregated at provider level to support the re-ranking of the Top-10 candidate providers into the final Top-5 recommendations.

Table 2.1:Comparison of existing techniques relevant to sentiment analysis and review credibility assessment21

Technique	**Main Purpose**	**Aspect-Level Analysis**	**Review Credibility**	**Main Limitation for This Research**
Numerical ratings	Overall provider evaluation	No	No	Does not represent detailed service-quality opinions
Traditional sentiment classification	Review sentiment classification	Limited/depends on approach	No	Strongly dependent on textual representation and does not assess review credibility
LSTM/Bi-LSTM sentiment analysis	Contextual sentiment classification	Possible	No	Sentiment evidence is not inherently credibility-aware
Attention-based	Focus on sentiment-relevant	Yes	No	Does not independently

sentiment analysis	textual information			evaluate review reliability
Multi-task sentiment analysis	Joint prediction of related sentiment tasks	Yes	No	Primarily addresses sentiment prediction
Supervised credibility detection	Classify suspicious/credible reviews	No	Yes	Requires reliable labelled training data
Proposed integrated approach	Aspect sentiment and credibility-informed provider evaluation	Yes	Yes	Designed specifically for review-based re-ranking of candidate providers

2.5 Gap synthesis and conceptual framework

The literature reviewed in the preceding sections demonstrates substantial progress in aspect-based sentiment analysis, deep sequential models, attention mechanisms, multi-task learning, review credibility assessment, and anomaly detection. These techniques provide effective mechanisms for extracting information from customer reviews; however, several limitations remain when they are considered in the context of service-provider evaluation and recommendation.

A major limitation identified in existing sentiment-analysis approaches is that sentiment prediction is commonly treated as the primary outcome of the analysis. Overall sentiment classification can represent the general polarity of a review but may conceal differences among individual dimensions of service performance. Aspect-Based Sentiment Analysis addresses this limitation by identifying sentiment associated with specific attributes. Nevertheless, aspect-level sentiment alone does not indicate whether the review providing the sentiment evidence should be considered equally reliable when compared with other reviews. Consequently, potentially anomalous or

low-credibility review evidence may still contribute to provider evaluation if credibility is not considered during subsequent processing.

Review credibility and suspicious-review detection research addresses a complementary problem. Existing approaches can analyze linguistic and behavioral characteristics to identify unusual review patterns, while unsupervised anomaly-detection techniques such as Isolation Forest provide an alternative when extensive labelled credibility data are unavailable. However, credibility assessment alone does not capture the opinions expressed by customers regarding individual service-quality dimensions. Therefore, neither aspect-level sentiment analysis nor review credibility assessment independently provides all the information required for a review-driven provider evaluation process.

A further gap concerns the transition from review-level analysis to provider-level decision making. Existing research frequently evaluates sentiment classification and credibility detection according to their respective predictive performance. However, for a service-provider recommendation system, review-level predictions must ultimately contribute to the evaluation of providers. This requires a mechanism for integrating aspect-level sentiment and review credibility, aggregating evidence from multiple reviews associated with the same provider, and using the resulting provider-level information to refine an existing recommendation ranking.

Based on these identified gaps, this research proposes the Bi-LSTM Based Sentiment Analysis and Review Credibility Module as an integrated review-analysis component. The conceptual framework begins with the Top-10 candidate service providers received from the preceding recommendation component. Customer reviews associated with these providers are analyzed through two complementary processes. First, a Multi-Task Bi-LSTM model with an Attention Mechanism predicts sentiment across Quality, Punctuality, Communication, and Professionalism. Second, an Isolation Forest-based credibility process analyses linguistic and behavioral review characteristics to provide an estimate of review credibility.

The outputs of these processes are subsequently integrated so that provider evaluation considers both the sentiment expressed in customer reviews and the estimated

credibility of the corresponding review evidence. The resulting review-level information is aggregated at provider level, and the derived provider scores are used to re-rank the initial Top-10 candidate providers. The five highest-ranked candidates are then returned as the final Top-5 service-provider recommendations. The conceptual framework therefore establishes a direct relationship between the identified literature gap and the proposed individual contribution by connecting aspect-level opinion modelling, review credibility assessment, evidence integration, and provider-level re-ranking within a unified processing pipeline.

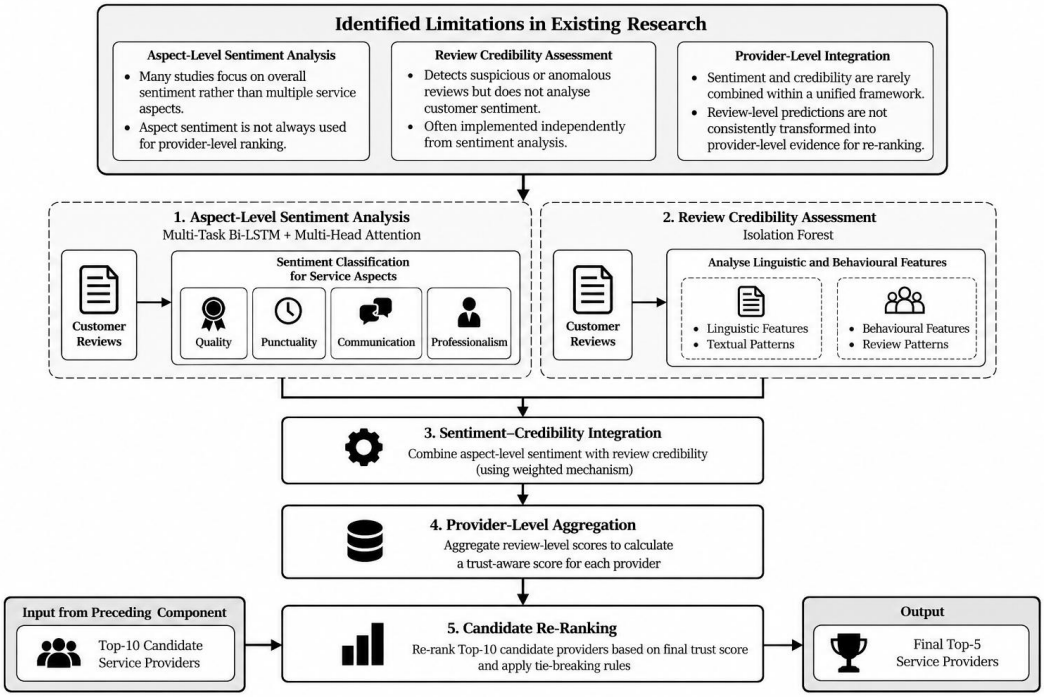


Figure 2.1:Conceptual framework derived from the identified research gap

2

3 RESEARCH METHODOLOGY

3.1 Research design

This research adopted a quantitative, experimental design to develop and evaluate the Bi-LSTM Based Sentiment Analysis and Review Credibility Module within the overall service-provider recommendation system. The methodology was designed to investigate two complementary forms of customer-review intelligence: aspect-level sentiment expressed regarding service quality and the estimated credibility of the review evidence. These outputs were subsequently integrated to support the provider re-ranking process.

The research design consisted of two parallel analytical processes followed by an integration and aggregation stage. The first process focused on aspect-level sentiment analysis. Customer reviews were preprocessed and used to train a Multi-Task Bidirectional Long Short-Term Memory (Bi-LSTM) model with an Attention Mechanism. The model was designed to predict sentiment independently for four service-quality aspects: Quality, Punctuality, Communication, and Professionalism. A multi-task architecture was selected so that the four related sentiment-classification tasks could learn from a shared representation of the review text while maintaining aspect-specific prediction outputs.

The second process focused on review credibility assessment. Linguistic and behavioral characteristics extracted from customer-review data were used as inputs to an Isolation Forest-based anomaly-detection model. The purpose of this stage was to identify reviews exhibiting unusual patterns and derive credibility-related information that could be considered together with the corresponding sentiment predictions. The anomaly output was therefore used as an analytical credibility signal rather than as definitive evidence that a review was genuine or fraudulent.

Following the two analytical processes, aspect-level sentiment information and review credibility information were combined within the trust-aware scoring stage. The resulting review-level evidence was aggregated for each service provider to obtain a provider-level trust-aware score. During system operation, the module receives the Top-10 candidate service providers generated by the preceding recommendation

component, analyses the reviews associated with these candidates, and uses the provider-level scores to re-rank the candidate set. The five highest-ranked candidates are subsequently returned as the final Top-5 service-provider recommendations.

The experimental evaluation was conducted separately for the sentiment-analysis and review-credibility models using quantitative performance measures. The sentiment model was evaluated for each of the four aspect-level classification tasks, while the credibility model was evaluated using its corresponding test data and classification measures. In addition, the integrated processing pipeline was examined to verify the transformation of review-level sentiment and credibility outputs into provider-level scores and the subsequent Top-10 to Top-5 re-ranking process.

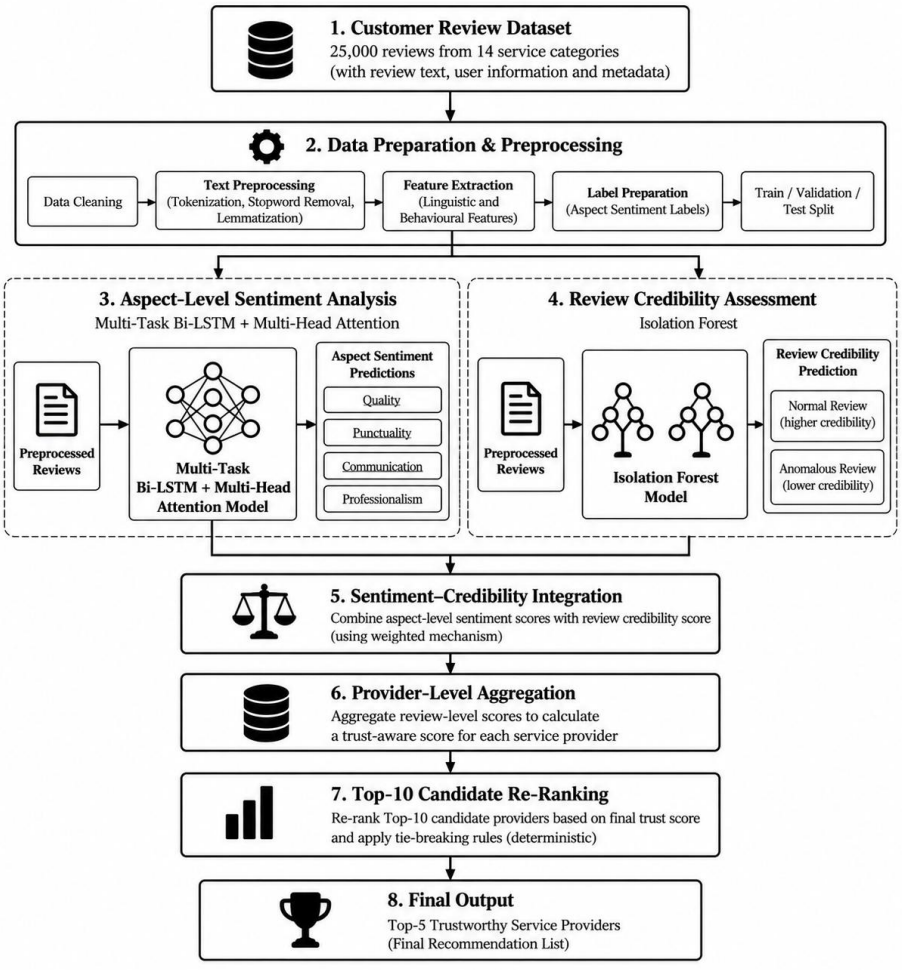


Figure 3.1:Research methodology of the Bi-LSTM Based Sentiment Analysis and Review Credibility Module

3

3.2 Data participants or study materials

The study materials used in this research consisted of an annotated customer-review dataset containing 25,000 reviews representing fourteen service categories. The dataset was prepared to support the development and evaluation of the Bi-LSTM Based Sentiment Analysis and Review Credibility Module. Each review record contained eleven attributes, including a unique review identifier, booking identifier, provider identifier, user identifier, service type, district, numerical rating, review text, review date, verified-booking status, and dataset-split information. Since the research was conducted as a computational study using customer-review records, no participants were directly recruited or involved in the experimental procedure.

The fourteen service categories represented in the dataset included A/C Technicians, Carpenters, Tile Workers, Cleaners, Cushion Works, Plumbers, Ceiling Works, Masons, Technicians, Well Services, CCTV Technicians, Electricians, Vehicle Repairs, and Painters. The distribution of reviews across these categories was maintained at a relatively comparable level, providing representation from different types of household and technical service providers rather than concentrating the analysis on a single service domain. This enabled the review-analysis models to be developed using customer feedback covering a diverse range of service contexts.

For the sentiment-analysis task, each review was associated with sentiment annotations for four service-quality aspects: Quality, Punctuality, Communication, and Professionalism. Each aspect was represented using three sentiment classes: Positive, Neutral, and Negative. The Quality aspect contained 11,972 Positive, 6,749 Neutral, and 6,279 Negative labels. Punctuality contained 12,042 Positive, 6,774 Neutral, and 6,184 Negative labels, while Communication contained 12,012 Positive, 6,862 Neutral, and 6,126 Negative labels. Professionalism contained 12,024 Positive, 6,767 Neutral, and 6,209 Negative labels. The similar class distributions observed across the four aspects provided a consistent basis for the multi-task sentiment-classification process.

At the overall sentiment level, Positive reviews represented approximately 47.0% of the dataset, while Neutral and Negative reviews represented approximately 26.9% and 26.1%, respectively. Therefore, although Positive sentiment formed the largest class,

both Neutral and Negative review instances were substantially represented within the dataset. Analysis of review length further indicated an average review length of approximately 16.8 words, with the dataset containing both shorter and longer customer-review statements. These characteristics were considered during the preparation of textual inputs for the sentiment-analysis model.

In addition to the sentiment annotations, a separate credibility dataset was prepared for the review credibility assessment process. The available credibility data contained the review-level information and derived characteristics required for identifying potentially anomalous review behavior. These data were used by the Isolation Forest-based credibility model to generate anomaly-related information for individual reviews. The resulting output was interpreted as an estimated credibility signal for subsequent integration with sentiment information rather than as definitive evidence that a review was genuine or fraudulent.

The dataset therefore provided two complementary forms of information required by the individual component. The annotated review text supported multi-task aspect-level sentiment analysis, while the credibility-related information supported anomaly-based review credibility assessment. The subsequent experimental procedure used these data to develop the two analytical processes before integrating their outputs for provider-level trust-aware evaluation.

3.3 Experimental or study procedure

The experimental procedure was conducted in sequential stages covering data preparation, dataset partitioning, aspect-level sentiment model development, review credibility assessment, integration of the two analytical outputs, and provider-level re-ranking. The two models were initially developed and evaluated as separate analytical processes before their outputs were combined within the Bi-LSTM Based Sentiment Analysis and Review Credibility Module.

The complete dataset consisted of 25,000 customer reviews. Following data preparation, the dataset was divided into training, validation, and test subsets. A total of 17,466 review instances were allocated to the training set, 3,761 instances to the validation set, and 3,773 instances to the test set. This corresponds approximately to

69.9% of the data for training, 15.0% for validation, and 15.1% for independent testing. The training subset was used for model parameter learning, the validation subset was used to monitor model behavior during training, and the test subset was retained for final performance evaluation.

3.3.1 Review data preparation

Customer-review records were first prepared for the two analytical processes. For sentiment analysis, the textual review content was converted into a representation suitable for sequential neural-network processing. Each review was associated with four aspect-specific target labels corresponding to Quality, Punctuality, Communication, and Professionalism. Each output task contained three possible sentiment classes: Positive, Neutral, and Negative. The prepared textual sequences and their corresponding aspect labels were subsequently supplied to the multi-task sentiment model.

A separate credibility representation was maintained for the review credibility process. The credibility dataset contained 25,000 review instances with the review-related characteristics required for anomaly-based assessment. These characteristics were transformed into structured model inputs before being supplied to the Isolation Forest algorithm. The sentiment and credibility data retained review-level correspondence so that the outputs generated by the two analytical processes could subsequently be integrated for the same review instance.

3.3.2 Multi-task Bi-LSTM sentiment model development

The aspect-level sentiment model was implemented as a multi-output neural architecture. The model accepted a review sequence with an input length of 80 tokens. The input was passed to an embedding layer that generated a 128-dimensional representation for each token. The resulting sequence was then processed using a Bidirectional Long Short-Term Memory layer, producing contextual representations that incorporated information from both directions of the review sequence.

An attention layer was applied to the Bi-LSTM representation to emphasize information relevant to sentiment classification. The resulting representation was subsequently passed through a shared dense layer containing 96 units followed by a

dropout stage. This shared representation was connected to four task-specific output layers corresponding to Quality, Punctuality, Communication, and Professionalism. Each output layer contained three units representing the Positive, Neutral, and Negative sentiment classes. This architecture enabled the four related sentiment tasks to share a common textual representation while maintaining independent aspect-level predictions.

The model was configured for a maximum of 20 training epochs. During training, performance on the validation data was monitored and the model state associated with improved validation loss was saved. This procedure allowed the best-performing model configuration observed during training to be retained for subsequent evaluation on the independent test set.

3.3.3 Review credibility model development

The second analytical process focused on estimating review credibility using Isolation Forest. The prepared credibility data were used to train the anomaly-detection model so that review instances exhibiting patterns substantially different from the dominant observations could be identified. Isolation Forest was selected as an unsupervised anomaly-detection mechanism, allowing anomalous patterns to be identified without relying exclusively on the sentiment information generated by the neural model.

Following model training, each review was processed to obtain anomaly-related information representing its credibility characteristics. Reviews following comparatively normal patterns were treated as having stronger credibility evidence, whereas anomalous observations were treated as having weaker credibility evidence. This output was retained as a credibility signal for the subsequent integration stage rather than being interpreted independently as conclusive evidence that a review was genuine or fraudulent.

3.3.4 Sentiment and credibility integration

After the two analytical processes were completed, the aspect-level sentiment outputs and review credibility information were associated at review level. The integration stage was designed to ensure that provider evaluation was influenced not only by the sentiment expressed within a customer review but also by the estimated credibility of

the corresponding review evidence. Consequently, review evidence with stronger credibility could contribute more reliably to subsequent provider-level evaluation, while evidence associated with anomalous review characteristics could be treated with reduced influence.

The integrated review-level information retained the four service-quality dimensions so that differences among Quality, Punctuality, Communication, and Professionalism were not lost during the combination process. The resulting trust-aware review evidence was then passed to the provider-level aggregation stage.

3.3.5 Provider-level aggregation and re-ranking

During operation within the complete recommendation system, the individual component receives the Top-10 candidate service providers generated by the preceding recommendation component. Customer reviews associated with these candidate providers are processed through the sentiment-analysis and credibility-assessment stages. The resulting review-level evidence is aggregated for each candidate provider to obtain a provider-level trust-aware score.

The calculated provider-level scores are then used to re-rank the initial Top-10 candidate set. Providers are ordered according to the resulting trust-aware evaluation, and the five highest-ranked candidates are returned as the final Top-5 service-provider recommendations. Therefore, the individual component functions as a review-intelligence and re-ranking stage rather than as an independent provider-retrieval mechanism.

3.4 Measures and evaluation metrics

The performance of the Bi-LSTM Based Sentiment Analysis and Review Credibility Module was evaluated using accuracy, precision, recall, and F1-score as the primary quantitative measures. These metrics were selected to evaluate both the aspect-level sentiment classification model and the review credibility assessment process from complementary perspectives.

For the Multi-Task Bi-LSTM with an Attention Mechanism, evaluation was conducted separately for the four sentiment-classification tasks: Quality, Punctuality, Communication, and Professionalism. Each task classified customer-review

information into Positive, Neutral, or Negative sentiment. Evaluating the four outputs separately enabled differences in classification performance among the service-quality aspects to be identified instead of representing the entire multi-task model using only a single overall score.

Accuracy was used to measure the overall proportion of correctly classified instances. Precision was used to determine the reliability of the model predictions, while recall measured the model's ability to correctly identify instances belonging to the corresponding sentiment classes. F1-score was considered together with precision and recall to provide a balanced assessment of classification performance. These measures were particularly useful for evaluating performance beyond overall accuracy when the distribution of sentiment classes was not completely uniform.

Confusion matrices were also used to examine the classification behaviour of the sentiment model across the Positive, Neutral, and Negative classes. This analysis provided additional information regarding correctly classified instances and the sentiment classes between which misclassification occurred. Separate confusion matrices were considered for Quality, Punctuality, Communication, and Professionalism.

The training process of the sentiment model was additionally assessed using training and validation accuracy and loss. These measures were monitored across the training epochs to examine model convergence and the relationship between training and validation performance. Validation loss was also monitored during model development to retain the model state associated with improved validation performance.

The Isolation Forest-based review credibility process was evaluated using accuracy, precision, recall, and F1-score for the resulting review classifications. Class-specific measures were considered for both normal and anomalous review instances. Macro-average and weighted-average performance measures were also examined because the two credibility classes were not equally represented in the test dataset. This enabled the credibility model to be assessed without relying exclusively on overall accuracy.

Collectively, these measures provided the basis for evaluating the two principal analytical processes of the proposed component. Aspect-specific measures were used to assess the Multi-Task Bi-LSTM with an Attention Mechanism, while class-specific and aggregate measures were used to assess the review credibility process. The numerical results, confusion matrices, training behavior, and corresponding interpretation are presented in the results and discussion chapter.

3.5 Data analysis

Data analysis was conducted at dataset, model, and integrated component levels to examine the behavior and performance of the Bi-LSTM Based Sentiment Analysis and Review Credibility Module. The analysis was based on the prepared customer-review dataset, the predictions generated by the Multi-Task Bi-LSTM with an Attention Mechanism, the outputs of the Isolation Forest-based credibility process, and the resulting provider-level trust-aware scores.

Initially, descriptive analysis was performed on the customer-review dataset to examine its overall composition before model evaluation. The distribution of reviews across the fourteen service categories was examined to determine the representation of different service domains. The distributions of Positive, Neutral, and Negative sentiment labels were also analyzed separately for Quality, Punctuality, Communication, and Professionalism. In addition, the overall sentiment distribution and review-length characteristics were examined to obtain a general understanding of the textual data used in the experiments. These analyses supported the interpretation of subsequent model results by providing information about class representation and review characteristics.

The performance of the Multi-Task Bi-LSTM with an Attention Mechanism was analyzed separately for each of the four aspect-level sentiment tasks. Accuracy, precision, recall, and F1-score were compared across Quality, Punctuality, Communication, and Professionalism to determine whether model performance differed according to the service-quality dimension being predicted. This aspect-specific analysis was important because the four classification tasks shared a common textual representation while producing independent sentiment outputs.

Training and validation behavior was examined across the model-training epochs using the recorded accuracy and loss values. The overall training and validation loss curves were analyzed to observe convergence during optimization, while aspect-specific training and validation accuracy curves were examined to determine whether improvements in training performance were reflected in validation performance. These observations were used to assess the stability of the learning process and to identify potential divergence between training and validation behavior.

The classification behavior of the sentiment model was further analyzed using normalized confusion matrices for each aspect. These matrices were examined to determine how effectively the model distinguished Positive, Neutral, and Negative sentiment and to identify the classes in which misclassification occurred. This provided a more detailed interpretation of model behavior than could be obtained from aggregate performance measures alone.

The review credibility process was analyzed separately using the predictions generated for the test dataset. Performance was examined for both normal and anomalous review classes using class-specific precision, recall, and F1-score together with overall, macro-average, and weighted-average measures. Particular attention was given to class-specific performance because the credibility test data contained different numbers of normal and anomalous review instances. This analysis provided a more balanced assessment of the credibility process than relying exclusively on overall accuracy.

Finally, the outputs of the sentiment and credibility processes were analyzed within the integrated component. Aspect-level sentiment information and credibility information were combined at review level and subsequently aggregated for each candidate service provider. The resulting provider-level scores were used to examine changes in the ordering of the initial Top-10 candidate providers. The analysis therefore considered not only the predictive performance of the individual models but also their role in supporting the final Top-5 trust-aware provider recommendation process.

3.6 Ethics privacy security and responsible AI

Ethical, privacy, security, and responsible Artificial Intelligence considerations were incorporated into the development of the Bi-LSTM Based Sentiment Analysis and Review Credibility Module. Since the component processes customer-review information and associated system records, particular attention was given to ensuring that review data were used only for purposes relevant to sentiment analysis, credibility assessment, and service-provider evaluation.

Privacy considerations were important because the review records are associated with users, bookings, and service providers within the overall system. The analytical models were designed to operate primarily on review text and the information required for the respective sentiment and credibility processes. Personally identifiable information was not required as an input to the sentiment-classification model. During model development and evaluation, review records were treated as analytical data rather than being used to identify or profile individual customers.

The integrated application uses Firebase for the storage and retrieval of system data required by the recommendation workflow. Access to review and provider information should therefore be restricted according to the permissions and authentication mechanisms of the application. Only information required by the corresponding system components should be retrieved and processed. This reduces unnecessary exposure of customer and provider information and supports appropriate control over data access within the integrated system.

Responsible interpretation was particularly important for the Isolation Forest-based review credibility process. The model identifies reviews that exhibit anomalous characteristics relative to patterns observed in the available data. An anomalous review cannot automatically be considered fraudulent, deceptive, or intentionally manipulated. Therefore, the credibility output is treated as an estimated reliability signal used to adjust the influence of review evidence during provider evaluation rather than as definitive proof that a reviewer has submitted a fake review. This distinction reduces the risk of making unsupported claims about individual users based solely on automated anomaly detection. [6],[9]

Potential bias was also considered in relation to both sentiment classification and credibility assessment. The performance of the Multi-Task Bi-LSTM with an Attention Mechanism depends on the characteristics and sentiment distributions represented in the training data. Similarly, the Isolation Forest model determines anomalous behavior relative to patterns contained in the available credibility data. Reviews containing uncommon language or unusual characteristics may therefore be treated differently even when they represent legitimate customer experiences. For this reason, model outputs are used as analytical signals within the recommendation process rather than as absolute judgements regarding customers or service providers.

The final provider re-ranking process was designed to combine multiple review observations rather than allowing a single review to determine the final recommendation outcome. Aspect-level sentiment information across Quality, Punctuality, Communication, and Professionalism is considered together with review credibility information and subsequently aggregated at provider level. This approach supports a more balanced use of customer-review evidence while recognizing that automated predictions may contain classification errors and uncertainty.

Overall, the component was developed with the principle that AI-generated sentiment predictions and credibility estimates should support the service-provider recommendation process rather than be interpreted as unquestionable decisions. Privacy-conscious data handling, controlled system access, cautious interpretation of anomaly detection, consideration of potential model bias, and aggregation of multiple review signals were therefore treated as important responsible-AI considerations throughout the research.

3.7 Reproducibility and research artefacts

Reproducibility was considered throughout the development of the Bi-LSTM Based Sentiment Analysis and Review Credibility Module by maintaining the principal data, model configurations, trained model artefacts, and evaluation outputs generated during the experimental process. These artefacts provide a record of the procedures followed during model development and support the replication or further extension of the individual research component.

The annotated customer-review dataset containing 25,000 reviews across fourteen service categories represents the primary research artefact used in this study. The dataset includes the review information required for aspect-level sentiment analysis and review credibility assessment. For sentiment analysis, annotations were maintained for Quality, Punctuality, Communication, and Professionalism using Positive, Neutral, and Negative sentiment classes. The prepared data were divided into 17,466 training instances, 3,761 validation instances, and 3,773 test instances, allowing the same data partitions to be distinguished throughout model development and evaluation.

The text-processing configuration was also maintained as part of the experimental setup. The sentiment-analysis pipeline used a vocabulary limit of 15,000 words and a maximum input sequence length of 80 tokens. The tokenizer was fitted using the training review text, and the same fitted tokenizer was subsequently applied to the validation and test data. This ensured that the three data partitions were processed using a consistent text representation while preventing the validation and test sets from being used to construct the training vocabulary.

The architecture and configuration of the sentiment model were recorded to support reproducibility. The implemented model consisted of an embedding layer with 128-dimensional representations, a Bidirectional LSTM layer with 96 units, a custom Attention Mechanism, a shared dense representation containing 96 units, and four task-specific output layers for Quality, Punctuality, Communication, and Professionalism. Each task-specific output performed three-class sentiment classification. The model was configured for a maximum of 20 training epochs, while validation performance was monitored during the training process.

Research artefacts were also maintained for the Isolation Forest-based review credibility process. These included the prepared credibility data, trained anomaly-detection model, generated credibility outputs, and corresponding evaluation results. Maintaining the sentiment and credibility artefacts separately enabled the behavior of the two analytical processes to be examined independently before their outputs were integrated within the complete component.

The experimental outputs generated during evaluation were retained as supporting research artefacts. These included training and validation histories, accuracy and loss curves, aspect-specific performance measures, normalized confusion matrices, credibility classification reports, and provider-level outputs generated during the integration and re-ranking stages. These materials provide supporting evidence for the quantitative findings presented in the results and discussion chapter.

Together, the documented dataset partitions, text-processing configuration, model architecture, trained model artefacts, and evaluation outputs provide a structured record of the experimental workflow. These artefacts support the reproducibility of the principal stages of the research and provide a foundation for future evaluation, modification, or extension of the Bi-LSTM Based Sentiment Analysis and Review Credibility Module.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The component requirements were identified according to its role within the complete service-provider recommendation system. The primary functional requirement was to analyses customer-review information associated with candidate service providers and produce review-based evidence that could be used to improve the final provider ranking. The component was therefore designed to receive candidate-provider information from the preceding recommendation component rather than independently searching the complete provider collection.

The sentiment-analysis functionality was required to process textual customer reviews and identify sentiment separately across four service-quality aspects: Quality, Punctuality, Communication, and Professionalism. For each aspect, the component was required to classify the corresponding sentiment into Positive, Neutral, or Negative categories. A shared multi-task architecture was required so that related information contained within a review could contribute to the four aspect-level prediction tasks while preserving separate outputs for each service-quality dimension.

A second functional requirement was the assessment of review credibility. The component was required to analyses the available credibility-related characteristics of individual reviews and identify potentially anomalous review patterns. The resulting anomaly information was required to be represented as a credibility signal that could subsequently influence the contribution of the corresponding review during provider evaluation.

The component was also required to integrate sentiment and credibility information at review level and aggregate the resulting evidence at provider level. During system operation, reviews associated with each of the Top-10 candidate providers are analyzed, and the resulting provider-level scores are used to re-rank the candidate set. The final functional output of the component is the Top-5 providers obtained after trust-aware review analysis and re-ranking.

In addition to these functional requirements, the component was designed with several non-functional considerations. The processing pipeline was required to apply

consistent text preparation and model inference procedures, preserve correspondence between reviews and their associated providers, and support integration with the overall application and Firebase-based data layer. The design also required credibility predictions to be interpreted as estimated reliability signals rather than definitive judgements regarding fraudulent behavior.

4.2 Component architecture within the complete solution

The Bi-LSTM Based Sentiment Analysis and Review Credibility Module operates as a downstream review-intelligence component within the complete service-provider recommendation solution. Its input is the Top-10 candidate provider set generated by the preceding recommendation component. The module retrieves or receives the customer-review information associated with these candidate providers and performs two parallel analytical processes.

The first processing path performs aspect-level sentiment analysis using the Multi-Task Bi-LSTM with an Attention Mechanism. Review text is transformed using the same text-processing configuration established during model development and passed to the trained sentiment model. The model generates separate sentiment outputs for Quality, Punctuality, Communication, and Professionalism.

The second processing path performs review credibility assessment using the Isolation Forest-based model. The required review characteristics are transformed into the representation expected by the credibility model, which generates anomaly-related information for each review. This information is subsequently represented as a credibility signal for the integration stage.

The outputs of the two processing paths are combined at review level before being aggregated according to provider. The resulting provider-level trust-aware scores are used to reorder the Top-10 candidate providers. The five highest-ranked candidates are then returned to the overall application as the final Top-5 recommendations. Firebase provides the data-storage layer required by the integrated application, while the review-intelligence component performs the analytical processing required for the final re-ranking stage.

4.3 Design alternatives and decisions

Several design alternatives were considered during the development of the individual component. One design decision concerned whether customer sentiment should be represented using a single overall sentiment prediction or multiple aspect-specific predictions. Overall sentiment provides a simplified representation of customer opinion but may conceal differences among individual dimensions of service performance. Therefore, the component was designed to produce separate sentiment predictions for Quality, Punctuality, Communication, and Professionalism.

A further decision concerned the sentiment-model architecture. Rather than developing four completely independent sentiment models, a multi-task architecture was adopted. The implemented Multi-Task Bi-LSTM enables the four sentiment tasks to learn from a shared textual representation while maintaining separate output layers. Bidirectional processing was used to capture contextual information from both directions of the review sequence, while the Attention Mechanism was incorporated to assign greater importance to relevant parts of the learned sequence representation.

For credibility assessment, an anomaly-detection approach was selected rather than making review credibility dependent solely on sentiment or numerical rating information. Isolation Forest was adopted to identify review observations exhibiting unusual patterns relative to the available data. This design also supports a cautious interpretation of the output as an anomaly-based credibility signal rather than a definitive fraud classification.

Another important design decision concerned the position of the component within the overall recommendation workflow. Instead of using review intelligence to search and rank every provider in the complete system, the component was designed as a re-ranking stage. It therefore receives the Top-10 candidate providers generated by the preceding recommendation component and applies the more detailed review analysis only to this candidate set. The final design consequently separates initial provider retrieval from trust-aware review-based evaluation.

Finally, the sentiment and credibility processes were designed as complementary analytical paths rather than replacing one another. Sentiment analysis determines what

customers express regarding the four service dimensions, whereas credibility assessment estimates the reliability of the corresponding review evidence. Integrating these outputs allows the final provider-level evaluation to consider both dimensions.

4.4 Detailed design

The detailed design consists of five principal processing stages: review preparation, aspect-level sentiment analysis, review credibility assessment, trust-aware integration, and provider-level aggregation and re-ranking.

The review-processing stage receives the customer reviews associated with the candidate service providers. For sentiment processing, the implemented tokenizer uses a vocabulary limit of 15,000 words, and review sequences are standardized to a maximum length of 80 tokens. The same text representation established during model development is applied during inference to maintain consistency between model training and operational processing.

The sentiment-analysis stage uses the trained Multi-Task Bi-LSTM with an Attention Mechanism. Each review sequence is passed through a 128-dimensional embedding layer followed by a Bidirectional LSTM containing 96 units. The Bi-LSTM produces sequential contextual representations that are subsequently processed by the custom Attention Mechanism. The resulting representation is passed through a shared dense layer containing 96 units with ReLU activation and a dropout layer. Four separate three-class softmax output layers then generate predictions for Quality, Punctuality, Communication, and Professionalism.

The credibility-assessment stage processes the corresponding credibility-related review characteristics using the trained Isolation Forest model. The resulting anomaly information is transformed into a credibility representation associated with the same review. Maintaining review-level correspondence between the sentiment and credibility paths enables their outputs to be integrated before provider-level aggregation.

The integration stage combines the aspect-level sentiment evidence with the corresponding review credibility information. This design prevents the provider evaluation process from treating sentiment predictions independently of the estimated

reliability of their underlying review evidence. The resulting review-level information is then grouped according to provider and aggregated to derive a trust-aware provider score.

The final design stage performs candidate re-ranking. Provider-level scores are associated with the Top-10 candidate providers received from the preceding component, and the candidates are ordered according to the resulting trust-aware evaluation. The five highest-ranked providers are selected and returned to the overall system as the final Top-5 service-provider recommendations.

4.5 Standards and professional practice

The design and implementation of the component followed established software-engineering and machine-learning practices to support maintainability, consistency, responsible use, and integration with the complete solution. The sentiment and credibility processes were implemented as logically separated analytical stages, allowing each model to be developed, evaluated, and maintained independently while exposing their outputs to a common integration process.

Consistent data-processing procedures were maintained between model development and operational inference. In particular, the tokenizer and sequence configuration established for the sentiment model were preserved so that new review text is represented in a manner consistent with the data used during training. Model configurations, trained artefacts, evaluation outputs, and dataset partitions were also maintained to support reproducibility and future modification of the component.

Professional practice was also considered in the handling of application data. The component was designed to process only the review and provider information required for its analytical functions, while Firebase serves as the data layer of the integrated system. Access to user-, review-, and provider-related information should remain subject to the authentication and access-control mechanisms defined by the complete application.

Responsible AI practice was particularly important for review credibility assessment. Anomaly detection identifies statistical deviations from learned patterns and does not independently establish fraudulent intent. Consequently, the credibility output is used

as a supporting signal within provider evaluation rather than as an automated accusation against an individual reviewer. Similarly, sentiment predictions are treated as model-generated estimates and are aggregated across available review evidence rather than allowing an individual prediction to determine the final provider recommendation. [6],[9]

The modular architecture, consistent processing pipeline, documented model configuration, privacy-conscious data handling, and cautious interpretation of AI-generated outputs were therefore incorporated as professional-practice considerations within the design of the component.

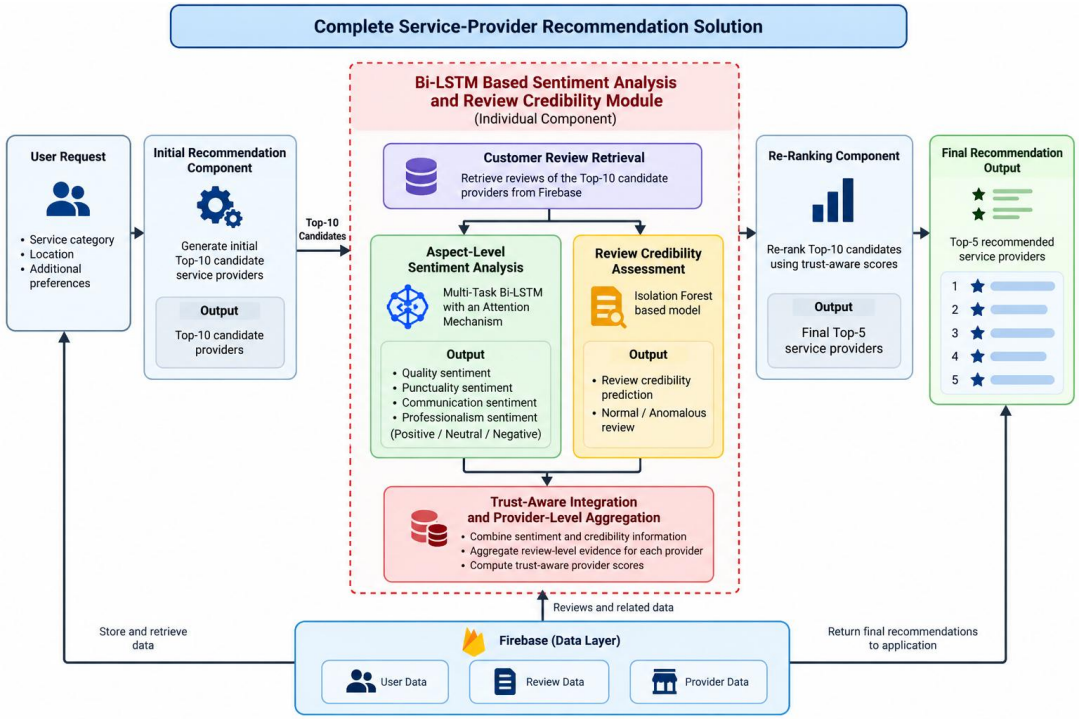


Figure 4.1:Detailed architecture of the Multi-Task Bi-LSTM with an Attention Mechanism

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The Bi-LSTM Based Sentiment Analysis and Review Credibility Module was implemented using a Python-based machine-learning environment. Google Colab was used during model development, training, and experimental execution, providing the computational environment required for data preparation and machine-learning model development. TensorFlow and Keras were used to construct and train the aspect-level sentiment classification model.

The sentiment-analysis implementation uses the Keras Functional API to construct a multi-task neural architecture. Customer-review text is transformed into numerical sequences using a tokenizer before being supplied to the model. The tokenizer was configured with a maximum vocabulary size of 15,000 words, while review sequences were standardized to a maximum length of 80 tokens. An out-of-vocabulary token was used to handle words that were not represented in the learned vocabulary.

The implemented sentiment model consists of an embedding layer, a Bidirectional Long Short-Term Memory layer, a custom Attention Mechanism, a shared dense representation, dropout regularization, and four task-specific classification outputs. The embedding layer generates 128-dimensional token representations, while the Bidirectional LSTM contains 96 units. The shared representation is subsequently used to generate sentiment predictions for Quality, Punctuality, Communication, and Professionalism.

The review credibility process was implemented using an Isolation Forest-based anomaly-detection approach. This process operates separately from the neural sentiment model and analyses the available review-related characteristics to produce credibility-related information for subsequent trust-aware processing.

Firebase is used as the application data layer within the integrated system. It supports the storage and retrieval of information required by the overall recommendation workflow, including provider and review information used by the individual component.

Table 5.1:Technologies used for component implementation51

Technology / Tool	**Purpose**
Python	Data processing and model implementation
TensorFlow / Keras	Multi-Task Bi-LSTM and Attention Mechanism implementation
Keras Tokenizer	Conversion of review text into numerical sequences
Sequence Padding	Standardization of review input length
Isolation Forest	Review credibility/anomaly assessment
Firebase	Application data storage and retrieval
Google Colab	Model development and experimental execution

5.2 Component implementation

The Bi-LSTM Based Sentiment Analysis and Review Credibility Module was implemented as two complementary analytical processes followed by an integration stage. The first process performs aspect-level sentiment analysis on customer-review text, while the second process performs review credibility assessment. The outputs generated by the two processes are subsequently associated with the corresponding reviews and prepared for provider-level trust-aware evaluation.

Customer-review text is first transformed into the representation expected by the sentiment model. The tokenizer converts the textual reviews into numerical token sequences using the vocabulary learned from the training data. These sequences are padded to a maximum length of 80 tokens to provide a consistent input shape for the neural model. The same text-processing configuration is maintained during training, validation, testing, and operational inference.

The sentiment-analysis process was implemented using a Multi-Task Bi-LSTM with an Attention Mechanism. Each token sequence is initially passed through a 128-dimensional embedding layer. The resulting sequence representation is processed by a Bidirectional LSTM containing 96 units. Bidirectional processing enables the model to construct contextual representations using information from both directions of the review sequence.

The sequential representations generated by the Bi-LSTM are subsequently supplied to a custom Attention Mechanism. The Attention Mechanism learns the relative importance of the generated sequence representations and produces a weighted representation of the customer review. This representation is then passed through a shared dense layer containing 96 units with ReLU activation. Dropout regularization is applied before the shared representation is supplied to the task-specific output layers.

Four independent output branches were implemented for Quality, Punctuality, Communication, and Professionalism. Each output layer uses softmax activation to produce a prediction across the three sentiment classes: Positive, Neutral, and Negative. The multi-task structure therefore enables a common representation of the customer review to support four related but independent aspect-level sentiment predictions.

The review credibility process was implemented separately using Isolation Forest. The required review-related characteristics are processed by the credibility model to identify observations that differ from the dominant patterns represented within the data. The resulting anomaly-related information is retained as a credibility signal associated with the corresponding review. This output is not independently treated as definitive evidence of fraudulent behavior.

Following the two analytical processes, the sentiment and credibility outputs are associated at review level. Each processed review therefore contains aspect-specific sentiment information together with corresponding credibility-related information. These outputs are subsequently used by the trust-aware processing stage for provider-level aggregation and candidate re-ranking.

5.3 Integration with the other components

The Bi-LSTM Based Sentiment Analysis and Review Credibility Module is integrated as a downstream review-intelligence component within the complete service-provider recommendation system. The module operates after the preceding recommendation component has generated an initial set of candidate service providers.

The preceding recommendation component provides the Top-10 candidate providers to the individual component. The relevant customer reviews associated with these providers are then obtained through the application data layer. This design prevents the individual component from independently searching and analyzing every provider available in the system and allows review intelligence to be applied specifically to an already identified candidate set.

The retrieved review text is processed through the trained Multi-Task Bi-LSTM with an Attention Mechanism to generate sentiment predictions for Quality, Punctuality, Communication, and Professionalism. In parallel, the required review-related information is processed by the Isolation Forest model to generate credibility-related evidence.

The outputs of the two analytical processes are associated using their corresponding review information before being passed to the trust-aware processing stage. Review-level evidence belonging to the same candidate provider is then aggregated to obtain provider-level information. This enables the system to evaluate each candidate using both the opinions expressed in customer reviews and the estimated credibility of the supporting review evidence.

The resulting provider-level trust-aware information is used to re-rank the initial Top-10 candidate set. Following re-ranking, the five highest-ranked providers are selected and returned to the overall application as the final Top-5 service-provider recommendations. Firebase supports the required data flow by providing application-side storage and retrieval of the relevant review and provider information.

This integration maintains a clear separation of responsibilities among the system components. Initial candidate selection is handled by the preceding recommendation component, while the Bi-LSTM Based Sentiment Analysis and Review Credibility

Module provides the additional review-intelligence layer required for trust-aware evaluation and final candidate re-ranking.

5.4 Verification and software testing

Verification and software testing were performed to confirm the correct operation of the individual analytical processes and their integration within the complete component. Testing was conducted for review preprocessing, aspect-level sentiment prediction, review credibility assessment, and the transfer of the resulting information into the provider-level processing stage.

The sentiment-analysis model was verified using training, validation, and independent test partitions. The training partition was used for model learning, while the validation partition was used to monitor model behavior during development. The independent test partition was retained for evaluating the trained model without using the same instances for model parameter learning.

Testing of the sentiment model was performed separately for the four output tasks: Quality, Punctuality, Communication, and Professionalism. The generated predictions were checked against the corresponding expected sentiment classes to verify that each output branch correctly operated as a three-class classification task representing Positive, Neutral, and Negative sentiment.

The model-training process was also monitored using training and validation behavior to verify that the learning procedure progressed as expected. Validation performance was monitored during training, and the model state associated with improved validation behavior was retained for subsequent evaluation. This allowed the selected model to be based on validation performance rather than simply using the state produced by the final training epoch.

The review credibility process was tested independently from the sentiment model. The designated test data were processed through the Isolation Forest-based credibility model to verify the generation of credibility-related outputs. Testing also considered the distinction between the dominant review patterns and review instances identified as comparatively anomalous.

Integration testing was subsequently performed to verify the transfer of information between the different stages of the component. The process was checked from the receipt of candidate-provider information through review retrieval, sentiment prediction, credibility assessment, review-level association, provider-level aggregation, and candidate re-ranking. Particular attention was given to maintaining the correspondence between each review, its analytical outputs, and the provider to which the review belonged.

The quantitative performance obtained from these verification and testing procedures, including model evaluation measures, training behavior, classification results, and graphical analyses, is presented separately in the Results and Discussion chapter.

5.5 Deployment and operation

The implemented component was prepared to operate as part of the integrated service-provider recommendation workflow. During normal operation, processing begins after the preceding recommendation component generates the Top-10 candidate service providers.

The component obtains the customer reviews associated with these candidate providers through the application data layer. Review text is processed using the same text-processing configuration used during model development. The reviews are converted into token sequences and standardized to the input length expected by the trained sentiment model.

The prepared review sequences are supplied to the trained Multi-Task Bi-LSTM with an Attention Mechanism. For each review, the model generates separate sentiment predictions for Quality, Punctuality, Communication, and Professionalism. In parallel, the required review-related information is supplied to the trained Isolation Forest model to generate credibility-related evidence.

The generated sentiment and credibility outputs are subsequently associated at review level and passed to the trust-aware processing stage. Review evidence belonging to each candidate service provider is aggregated to produce provider-level information. This information is then used to re-rank the initial Top-10 candidate providers.

Following the re-ranking process, the five highest-ranked providers are selected as the final Top-5 recommendations and returned to the overall application. The operational workflow therefore allows the component to function as a specialized review-intelligence and re-ranking stage rather than as an independent provider-retrieval mechanism.

Firebase supports the operational workflow by providing the application-side storage and retrieval required for provider and review information. The modular implementation of sentiment analysis, credibility assessment, and provider-level processing also allows the individual analytical stages to be maintained or updated without changing the fundamental role of the component within the complete recommendation system.

6 RESULTS AND VALIDATION

6.1 Research results

The experimental results demonstrate the performance of the proposed Bi-LSTM Based Sentiment Analysis and Review Credibility Module in analyzing customer reviews at both sentiment and credibility levels. The research evaluation focused on the four aspect-level sentiment classification tasks, namely Quality, Punctuality, Communication, and Professionalism, together with the review credibility assessment performed using the Isolation Forest model.

The Multi-Task Bi-LSTM with an Attention Mechanism demonstrated consistent performance across the four sentiment-analysis tasks. Among the evaluated aspects, Quality achieved the highest classification performance, followed by Punctuality, Communication, and Professionalism. The average test accuracy across all four sentiment-classification tasks was 88.94%.

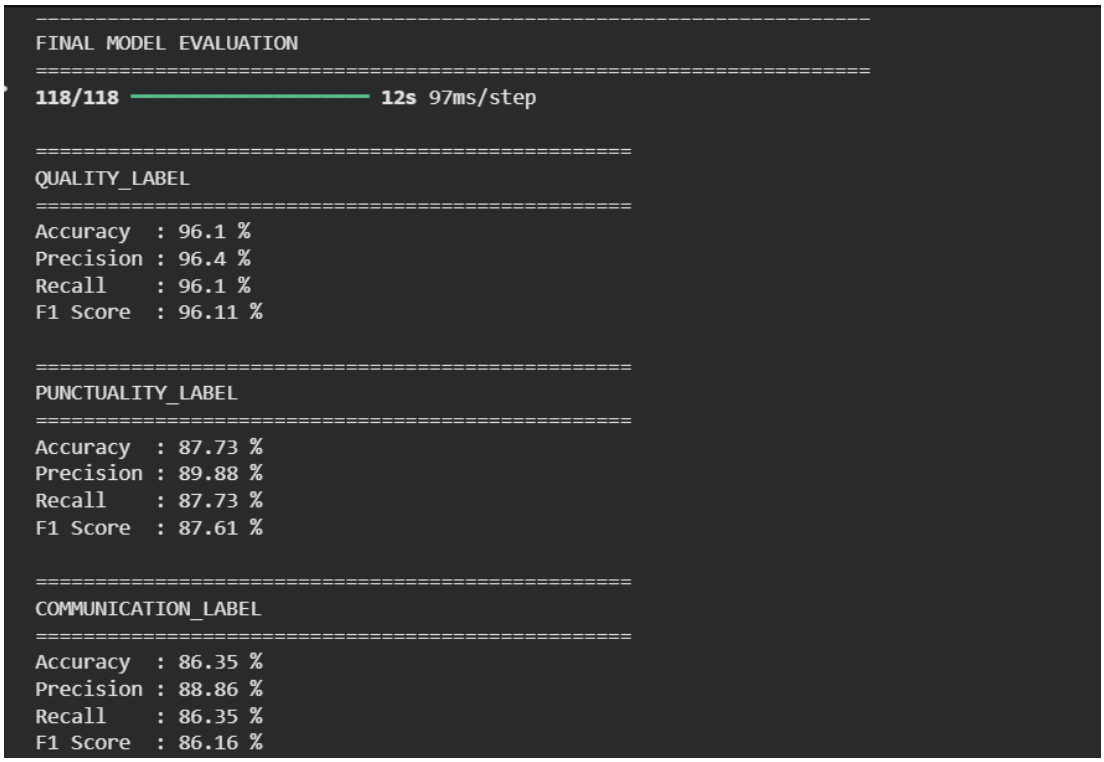


Figure 6.1:Final evaluation results of the aspect-level sentiment classification model

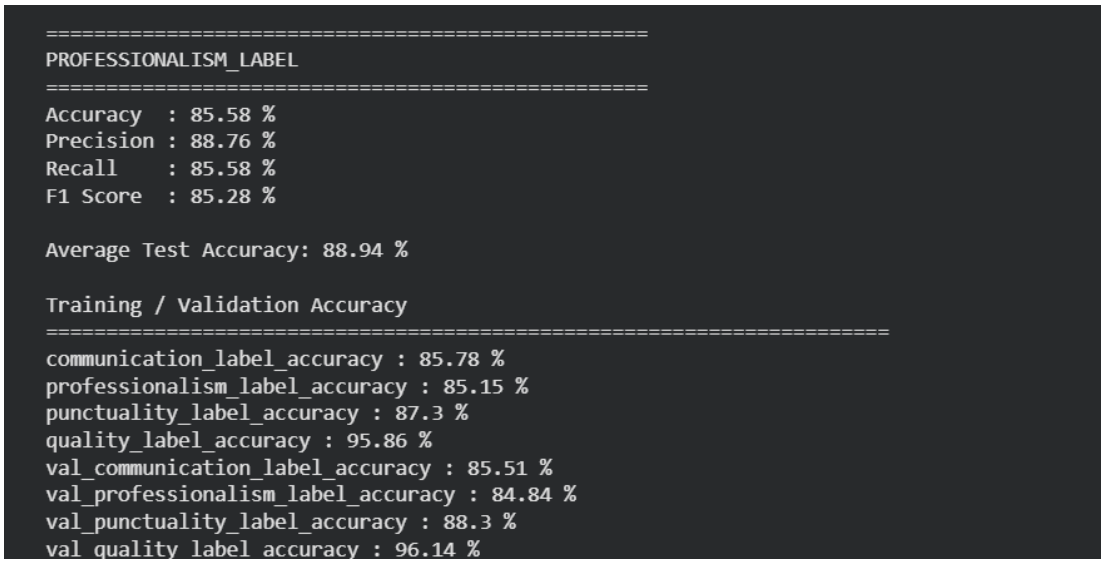


Figure 6.2:Final evaluation results for Professionalism sentiment classification

Table 6.1:Performance of the Aspect-Level Sentiment Classification Model61

Aspect	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1-Score (%)**
Quality	96.10	96.40	96.10	96.11
Punctuality	87.73	89.88	87.73	87.61
Communication	86.35	88.86	86.35	86.16
Professionalism	85.58	88.76	85.58	85.28

The Quality aspect achieved the highest accuracy of 96.10% with an F1-score of 96.11%. Punctuality achieved an accuracy of 87.73%, while Communication and Professionalism achieved accuracies of 86.35% and 85.58%, respectively. The average test accuracy across the four classification tasks was 88.94%.

Training behavior

The training and validation behavior of the model was examined to determine whether the learning process remained stable during model development.

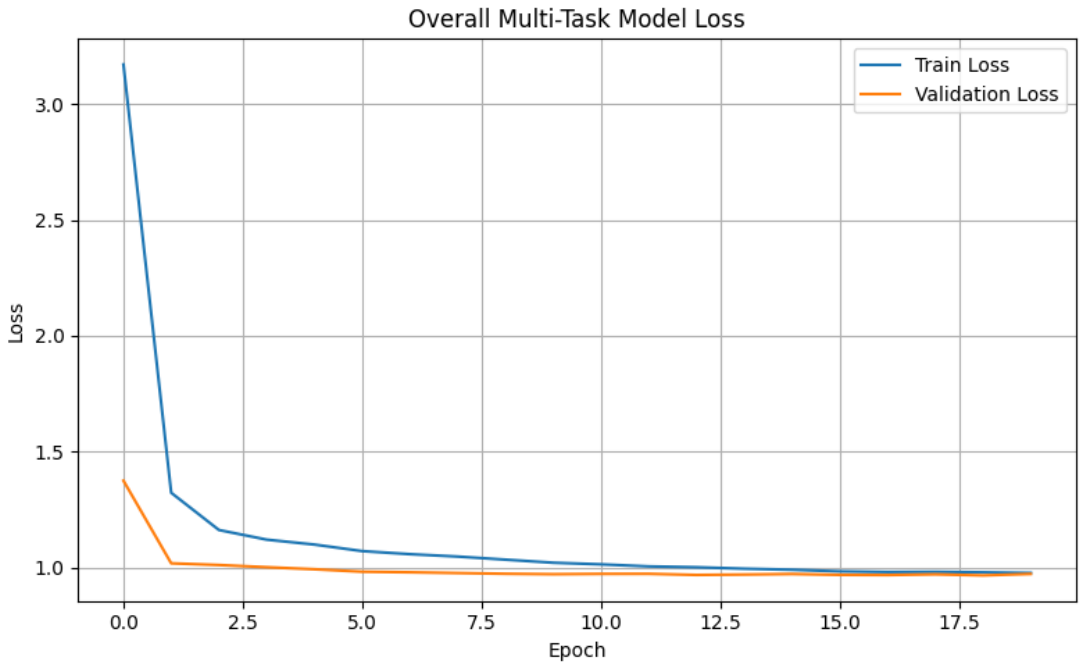


Figure 6.3:Overall training and validation loss of the Multi-Task Bi-LSTM model

The overall loss curves show a substantial reduction during the initial training epochs, followed by a comparatively stable pattern during the later stages. The validation-loss behavior remained relatively close to the training behavior, providing evidence that the model reached a stable learning state.

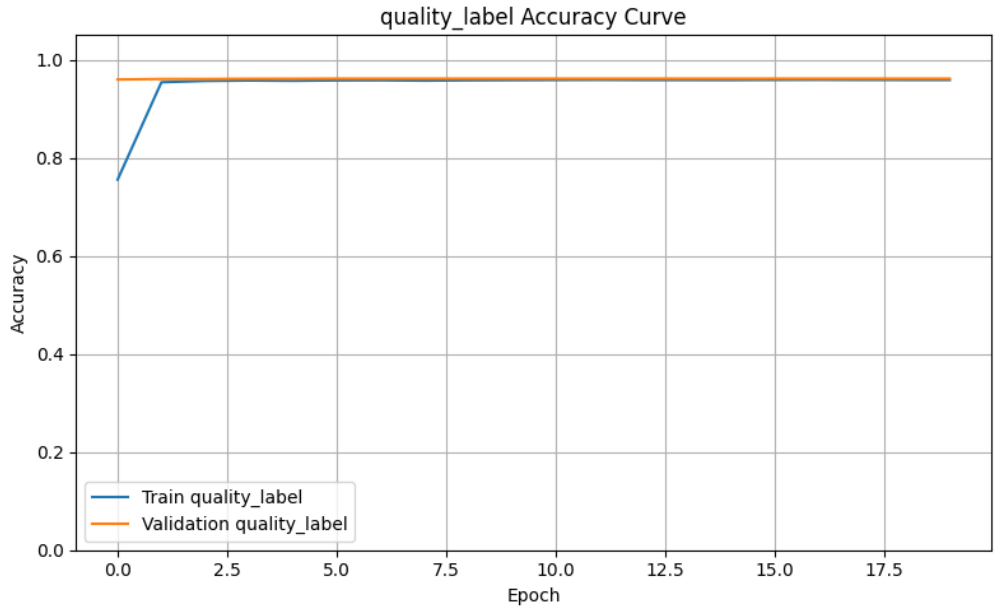


Figure 6.4:Training and validation accuracy for Quality sentiment classification

The Quality sentiment task demonstrated rapid improvement during the initial training stages. Training and validation accuracy subsequently remained close to each other and stabilized at approximately 96%, which is consistent with the strong final test performance obtained for this aspect. 6

6.2 Product and technical results

The technical results were further examined using the individual classification behavior of the four aspect-level sentiment tasks. The normalized confusion matrices provide a detailed representation of correct classifications and misclassifications among Positive, Neutral, and Negative sentiment classes.

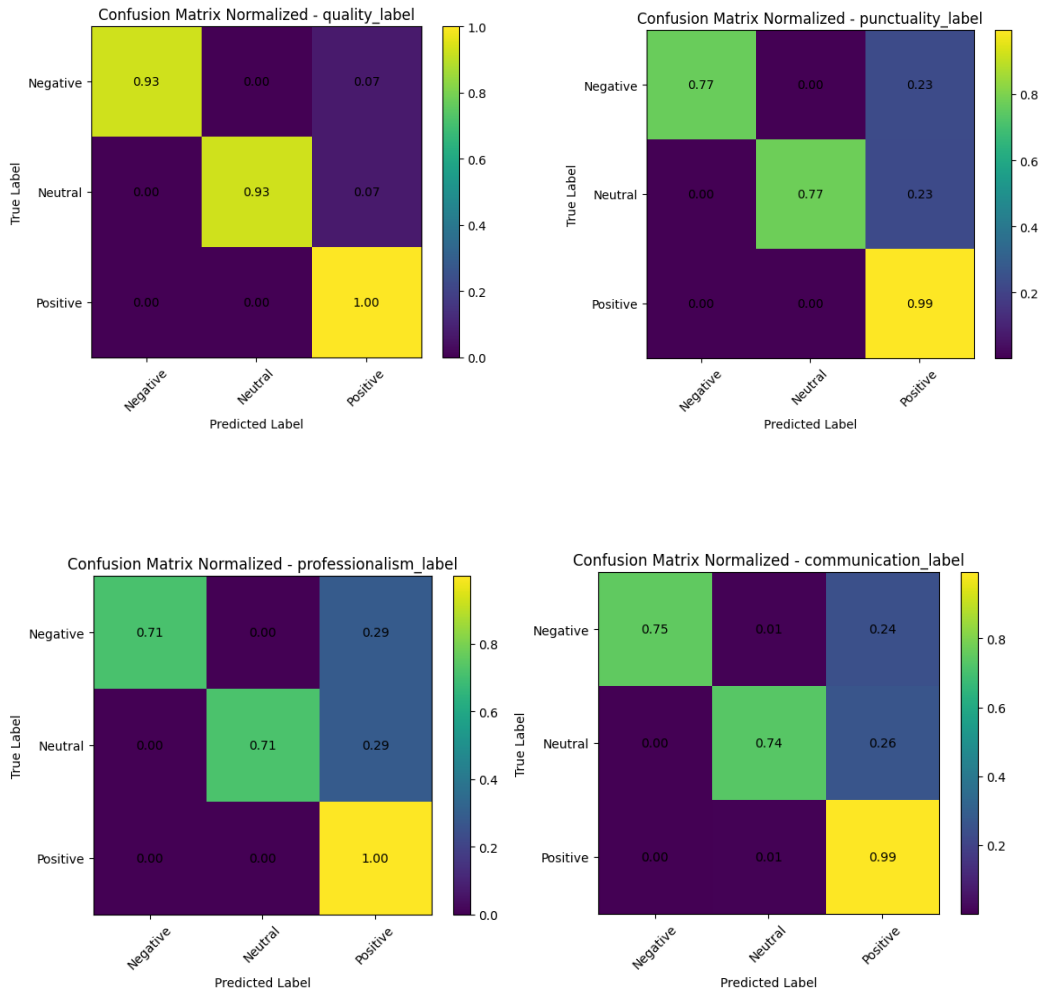


Figure 6.5:Normalized confusion matrices for the four aspect-level sentiment classification tasks

The confusion matrices demonstrate particularly strong recognition of Positive sentiment across the four aspects. Quality showed the strongest overall separation among the sentiment classes. Comparatively greater confusion was observed for Negative and Neutral sentiment in Punctuality, Communication, and Professionalism. This observation is consistent with the lower overall classification performance obtained for these aspects.

Review credibility results

The review credibility component was evaluated independently to determine its ability to identify normal and anomalous review patterns.

```
==========================================================================
Test Fake Review Detection Report
==========================================================================
              precision    recall  f1-score   support

        Real       0.98      0.97      0.98      3387
        Fake       0.77      0.80      0.79       386

    accuracy                           0.96      3773
   macro avg       0.87      0.89      0.88      3773
weighted avg       0.96      0.96      0.96      3773
```

Figure 6.6:Evaluation results of the Isolation Forest-based review credibility model

Table 6.2:Performance of the Review Credibility Model62

Review Class	Precision	Recall	F1-Score	Support
Real	0.98	0.97	0.98	3387
Fake	0.77	0.80	0.79	386
Macro Average	0.87	0.89	0.88	3773
Weighted Average	0.96	0.96	0.96	3773

The credibility model achieved an overall accuracy of 96%. The Real class achieved an F1-score of 0.98, while the Fake class achieved an F1-score of 0.79. The weighted-average precision, recall, and F1-score were each 0.96. The results indicate that anomalous review instances represented a more challenging case than reviews exhibiting normal patterns.

6.3 User or stakeholder validation

The validation of the individual component primarily focused on technical and functional evaluation. The implemented component was examined in terms of its ability to analyze customer reviews, generate aspect-level sentiment predictions,

assess review credibility, and provide the resulting information for trust-aware provider evaluation.

The component was designed so that the machine-learning outputs contribute to the final provider re-ranking process rather than being presented to users as isolated predictions. The sentiment and credibility information therefore functions as supporting evidence for the overall recommendation process.

A broader end-user or stakeholder study could be conducted as future work to evaluate perceived recommendation quality, usability, trust, and user satisfaction with the complete integrated system.

6.4 Integration results

The integration evaluation considered whether the individual component could successfully operate as part of the complete service-provider recommendation workflow. The process begins when the preceding recommendation component supplies the Top-10 candidate service providers.

The relevant review information for these candidates is processed by the sentiment-analysis and credibility-assessment models. Aspect-level sentiment information and credibility-related evidence are subsequently associated with the relevant reviews and aggregated at provider level. The resulting trust-aware information supports the re-ranking of the initial candidate set and selection of the final Top-5 service providers.

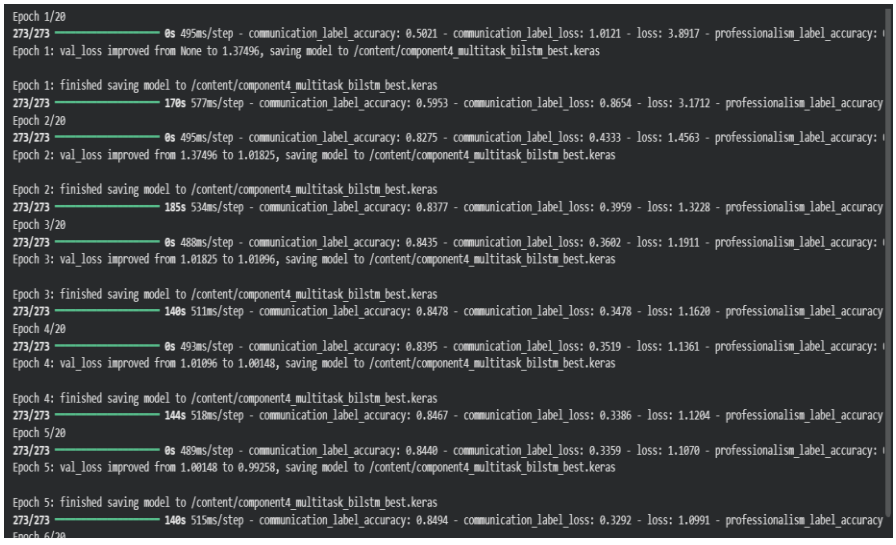


Figure 6.7:Training progress of the Multi-Task Bi-LSTM sentiment model

6.5 Summary of findings

The experimental results demonstrate that the Bi-LSTM Based Sentiment Analysis and Review Credibility Module achieved effective performance across its two primary analytical tasks. The Multi-Task Bi-LSTM with an Attention Mechanism achieved an average test accuracy of 88.94% across Quality, Punctuality, Communication, and Professionalism. Quality produced the highest individual accuracy of 96.10%, while the remaining aspects also achieved accuracies above 85%.

The training and validation results demonstrated stable learning behavior, while the detailed classification analysis showed stronger recognition of Positive sentiment compared with some Negative and Neutral instances. The review credibility model achieved an overall test accuracy of 96%, providing an additional credibility-related signal for the review-analysis process.

The integration of sentiment and credibility information enables review evidence to be considered at provider level. This information supports the trust-aware re-ranking of the Top-10 candidate service providers and contributes to the generation of the final Top-5 recommendations.

7 DISCUSSION

7.1 Interpretation of findings

The findings demonstrate that the proposed Bi-LSTM Based Sentiment Analysis and Review Credibility Module can extract useful review intelligence from customer feedback by considering both aspect-level sentiment and review credibility. Rather than treating a customer review as a single overall opinion, the implemented approach evaluates sentiment separately for Quality, Punctuality, Communication, and Professionalism. This provides a more detailed representation of customer experience that can subsequently support provider-level evaluation.

The Multi-Task Bi-LSTM with an Attention Mechanism achieved an average test accuracy of 88.94% across the four aspect-level sentiment classification tasks. However, the results were not identical across all aspects. Quality achieved the strongest performance with an accuracy of 96.10%, followed by Punctuality at 87.73%, Communication at 86.35%, and Professionalism at 85.58%. This variation suggests that the linguistic patterns associated with Quality were more clearly distinguishable within the available review data than those associated with Communication and Professionalism.

The normalized confusion matrices provide further insight into this behavior. For Quality, the model correctly classified 93% of both Negative and Neutral instances and 100% of Positive instances. Positive sentiment was also strongly recognized for Punctuality and Communication at 99%, while Professionalism achieved 100% correct classification for the Positive class. In comparison, Negative and Neutral sentiment showed lower correct classification rates for Punctuality, Communication, and Professionalism.

A notable pattern was the tendency for some Negative and Neutral instances to be classified as Positive. This behavior was particularly visible for Professionalism, where approximately 29% of both Negative and Neutral instances were classified as Positive, and for Communication, where approximately 24% of Negative and 26% of Neutral instances were classified as Positive. This indicates that distinguishing less

strongly expressed sentiment may be more difficult for the model than recognizing clearly Positive review expressions.

The credibility analysis provides a second perspective on the review evidence. The Isolation Forest-based process achieved an overall test accuracy of 96%, with stronger performance for the Real class than the Fake class. The Fake class achieved a recall of 0.80 and an F1-score of 0.79. This difference indicates that anomalous review patterns are more difficult to distinguish than dominant review patterns. Consequently, the credibility output is most appropriately interpreted as supporting evidence of review reliability rather than definitive proof that a review is fraudulent.

Taken together, the findings support the main design principle of the component: sentiment information alone does not provide a complete representation of review evidence. Aspect-level sentiment explains what customers express about different characteristics of a service, while credibility assessment provides an additional indication of how the review evidence should be treated. Their combination therefore provides a stronger basis for trust-aware provider evaluation than relying only on overall ratings or a single review-level sentiment label.

7.2 Comparison with literature baselines and existing solutions

The proposed component differs from conventional service-review analysis approaches that primarily depend on overall star ratings, aggregated review scores, or general positive and negative sentiment. Such approaches provide useful high-level information but may not represent the different dimensions of service quality that influence a customer's experience. A provider may, for example, receive different opinions regarding service quality, punctuality, communication, and professionalism even when the overall rating is similar. [1],[5]

The implemented approach addresses this issue through multi-task aspect-level sentiment classification. A shared Bi-LSTM representation is used to support four related sentiment-classification tasks, while the Attention Mechanism enables the model to assign greater importance to relevant sequential information within each review. This differs from a single-output sentiment classifier because the same

customer review can contribute separate sentiment evidence for multiple service-related aspects. [2],[4]

A further distinction is the inclusion of review credibility assessment. Many sentiment-oriented solutions focus on the content and polarity of customer opinions without considering whether the review itself exhibits unusual characteristics. In the proposed component, Isolation Forest-based anomaly detection provides an additional credibility-related signal. The provider evaluation process can therefore consider both the expressed customer opinion and the credibility-related evidence associated with the review. [6],[9]

The achieved experimental performance demonstrates the feasibility of this combined approach. The aspect-level sentiment model achieved an average accuracy of 88.94%, while the credibility model achieved an overall accuracy of 96%. However, direct numerical comparison with previously published models must be interpreted carefully because differences in datasets, service domains, sentiment classes, aspect definitions, class distributions, preprocessing procedures, and evaluation protocols can substantially affect reported performance. [1],[6]

Therefore, the main comparison with existing approaches is not limited to obtaining a higher numerical accuracy. The principal distinction of the proposed approach is the integration of multi-aspect sentiment information and review credibility within a service-provider recommendation workflow, allowing review evidence to contribute to trust-aware re-ranking of an initially recommended candidate set. [10],[11]

7.3 Research contribution

The primary research contribution of this study is the development and evaluation of a review-intelligence approach that combines aspect-level sentiment analysis with review credibility assessment for service-provider recommendation. The research moves beyond the use of a single overall review sentiment by representing customer opinion through four service-specific aspects: Quality, Punctuality, Communication, and Professionalism.

A second contribution is the application of a multi-task learning structure to the aspect-level sentiment problem. Instead of developing four completely independent sentiment

models, the implemented Multi-Task Bi-LSTM uses a shared representation of review text and produces four task-specific outputs. This enables common linguistic information within customer reviews to contribute to multiple related aspect-classification tasks while maintaining separate sentiment predictions for each aspect.

The custom Attention Mechanism provides an additional contribution by allowing the model to assign different levels of importance to the sequential representations generated by the Bidirectional LSTM. This supports the extraction of a weighted review representation before aspect-specific classification.

The research further contributes by considering review credibility alongside sentiment information. Isolation Forest-based anomaly detection was incorporated as a separate analytical process to provide credibility-related evidence. This enables the research to consider not only the sentiment contained within a review but also whether the review exhibits patterns that differ from the dominant review behavior.

Finally, the research demonstrates how the outputs of these analytical processes can support a trust-aware provider evaluation workflow. The component operates on the Top-10 candidates generated by the preceding recommendation component and provides review-derived information that supports subsequent provider-level aggregation and re-ranking. This establishes a connection between review-level machine-learning analysis and provider-level recommendation decisions.

7.4 Product contribution

The product contribution of the individual component is its role as a specialized review-intelligence layer within the complete service-provider recommendation system. The component enhances the initial recommendation process by analyzing the customer-review evidence associated with the candidate providers rather than independently generating candidates from the complete provider population.

The preceding recommendation component initially produces the Top-10 candidate service providers. The individual component then processes the reviews associated with these candidates. The Multi-Task Bi-LSTM with an Attention Mechanism generates sentiment information for Quality, Punctuality, Communication, and

Professionalism, while the Isolation Forest model provides credibility-related evidence.

This design provides the overall product with more detailed information than would be available from overall ratings alone. For example, two candidate providers with similar general ratings may have different patterns of customer sentiment regarding punctuality or communication. Aspect-level analysis allows these differences to be represented within the review-intelligence process.

The credibility process also contributes to the product by providing an additional signal when review evidence is evaluated. Rather than assuming that every review should automatically contribute equally to provider assessment, credibility-related information can be considered alongside the sentiment output during trust-aware processing.

The individual component therefore contributes directly to the transition from the initial Top-10 candidate set to the final Top-5 service-provider recommendations. Its product-level contribution is not the replacement of the original recommendation mechanism, but the addition of a review-based analytical stage that supports more informed and trust-aware candidate re-ranking.

7.5 Limitations and threats to validity

Several limitations should be considered when interpreting the findings of this research. First, the performance of the sentiment model depends on the characteristics and representativeness of the available dataset. Although the dataset contains 25,000 reviews distributed across 14 service categories, the linguistic patterns represented in the dataset may not capture every form of customer expression that could occur in real-world service reviews.

A second limitation concerns differences in performance among the four sentiment aspects. Quality achieved substantially stronger classification performance than Communication and Professionalism. The confusion-matrix analysis also demonstrated that some Negative and Neutral instances were classified as Positive, particularly for Communication and Professionalism. Therefore, the model should not be assumed to provide identical reliability across every sentiment aspect and class.

The credibility process also has limitations. Isolation Forest identifies anomalous patterns relative to the broader data distribution, but an anomalous review is not necessarily fraudulent. Similarly, a review that follows common patterns cannot automatically be considered completely trustworthy. For this reason, credibility outputs should be interpreted as supporting signals rather than definitive fraud labels. [6],[9]

Class distribution represents another consideration. The credibility test data contained considerably more Real instances than Fake instances, with 3,387 Real reviews compared with 386 Fake reviews. Overall accuracy alone could therefore provide an incomplete representation of credibility-model performance. Class-specific precision, recall, and F1-score are important when interpreting these results.

The study also primarily validates the individual component through experimental and technical evaluation. A large-scale end-user or stakeholder validation study was not used to establish perceived trust, usability, or satisfaction with the final recommendation output. Consequently, technical model performance should not automatically be interpreted as evidence of improved user satisfaction.

Another threat to external validity is that the evaluation was performed within the service categories and data characteristics represented by the current study. Performance may change when the model is applied to substantially different service domains, writing styles, vocabulary, or future review data. Further evaluation using independently collected real-world review datasets would therefore strengthen evidence of generalizability.

Finally, the final provider recommendation depends on integration with other system components. Strong sentiment and credibility performance does not independently guarantee that every final Top-5 recommendation will be optimal because the final output is also influenced by the initial Top-10 candidate set and the subsequent provider-level integration and re-ranking process.

7.6 Implications

The findings have implications for both review analytics and service-provider recommendation systems. At the analytical level, the results demonstrate the value of

examining customer feedback at aspect level rather than representing an entire review through a single sentiment label. Service-provider evaluation can benefit from distinguishing between Quality, Punctuality, Communication, and Professionalism because these dimensions represent different characteristics of the service experience. [1],[5]

The findings also demonstrate the importance of considering the credibility of review evidence. Customer reviews can strongly influence recommendation decisions, and treating all available reviews as equally reliable may reduce the quality of the evidence used by the system. Incorporating anomaly-based credibility information provides an additional mechanism for considering the reliability of review evidence during provider evaluation. [6],[10]

For recommendation-system design, the research demonstrates that review intelligence can be introduced as a downstream re-ranking stage. The component does not need to replace the initial recommendation mechanism. Instead, an initial candidate set can be generated first and then refined using more computationally detailed sentiment and credibility analysis. In this study, this principle is applied by analyzing the reviews associated with the Top-10 candidates before selecting the final Top-5 providers.

From a practical perspective, the approach could support more informative service-provider selection because the recommendation process can consider multiple dimensions of previous customer experience together with credibility-related information. However, the analytical outputs should remain supporting signals rather than absolute judgements, particularly in relation to review credibility.

The findings also identify opportunities for future improvement. Additional real-world review data could be used to evaluate generalizability, while further investigation could focus on improving the distinction between Negative and Neutral sentiment for the more challenging aspects. Broader stakeholder validation could also determine whether incorporating trust-aware review intelligence improves user confidence and perceived usefulness of the final recommendations.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This research developed a Bi-LSTM Based Sentiment Analysis and Review Credibility Module to support trust-aware service-provider recommendation. The component combines aspect-level sentiment analysis with review credibility assessment to analyze customer-review evidence associated with candidate service providers.

The Multi-Task Bi-LSTM with an Attention Mechanism was used to classify customer sentiment across Quality, Punctuality, Communication, and Professionalism, while Isolation Forest was used to provide credibility-related information by identifying anomalous review patterns. The sentiment model achieved an average test accuracy of 88.94%, while the credibility model achieved an overall accuracy of 96%.

The developed component operates after the initial recommendation process by analyzing reviews associated with the Top-10 candidate providers. The resulting sentiment and credibility information supports trust-aware provider re-ranking and the selection of the final Top-5 service providers. Overall, the research demonstrates the feasibility of combining detailed sentiment information with review credibility assessment within a service-provider recommendation workflow.

8.2 Achievement of objectives

The objectives defined for the individual research component were successfully addressed through the development and evaluation of the proposed solution. Aspect-level sentiment analysis was implemented to classify customer opinions across four service-related aspects: Quality, Punctuality, Communication, and Professionalism.

A separate review credibility process was implemented using Isolation Forest to provide anomaly-based credibility information. The outputs of the sentiment and credibility processes were subsequently incorporated into a trust-aware provider evaluation workflow.

The developed component was also integrated with the overall recommendation process, enabling the initial Top-10 candidate providers to be evaluated using their

associated review evidence before producing the final Top-5 recommendations. The experimental evaluation confirmed the technical feasibility of the implemented approaches.

8.3 Individual research and product contributions

The main research contribution is the combination of multi-task aspect-level sentiment analysis and review credibility assessment for service-provider evaluation. The Multi-Task Bi-LSTM with an Attention Mechanism provides separate sentiment predictions for four service aspects, while the Isolation Forest model provides an additional credibility-related signal.

From a product perspective, the component provides a review-intelligence and trust-aware re-ranking stage for the overall recommendation system. Instead of replacing the initial recommendation process, it enhances the candidate-selection workflow by analyzing the review evidence associated with the Top-10 candidates and supporting the selection of the final Top-5 providers.

8.4 Future directions

Future work can focus on improving sentiment classification for the more challenging Communication and Professionalism aspects, particularly the distinction between Negative and Neutral sentiment. Additional real-world review data from different service categories could also be used to improve model generalizability.

The credibility assessment could be extended by investigating additional review characteristics and alternative anomaly-detection or credibility-modelling approaches. Future research could also evaluate the complete recommendation workflow with a larger group of real users and stakeholders to measure usability, perceived trust, and recommendation satisfaction.

Finally, continuous model updating could be investigated so that the system can adapt to newly collected reviews and changing customer language over time.

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10 GLOSSARY

Aspect-Level Sentiment Analysis – The process of identifying sentiment separately for specific aspects of a service, such as Quality, Punctuality, Communication, and Professionalism.

Attention Mechanism – A neural-network mechanism that assigns different levels of importance to sequential information when generating a representation of a customer review.

Bidirectional LSTM – A sequence-processing neural-network structure that processes textual information in both forward and backward directions to capture contextual information.

Multi-Task Learning – A machine-learning approach in which a shared model representation is used to perform multiple related prediction tasks. In this research, it is used for the four aspect-level sentiment classification tasks.

Review Credibility – The degree to which review evidence is considered reliable for use within the trust-aware provider evaluation process.

Anomaly Detection – The process of identifying data instances that exhibit unusual patterns compared with the majority of the dataset.

Isolation Forest – An anomaly-detection technique used in this research to identify unusual review patterns for credibility assessment.

Trust-Aware Evaluation – The evaluation of candidate service providers using review-derived sentiment information together with credibility-related evidence.

Provider-Level Aggregation – The process of combining review-level analytical outputs associated with the same service provider to obtain provider-level information.

Candidate Re-ranking – The process of changing the ordering of initially recommended service providers using additional analytical information.

Service Provider – An individual or entity offering a service to customers through the service-provider recommendation system.

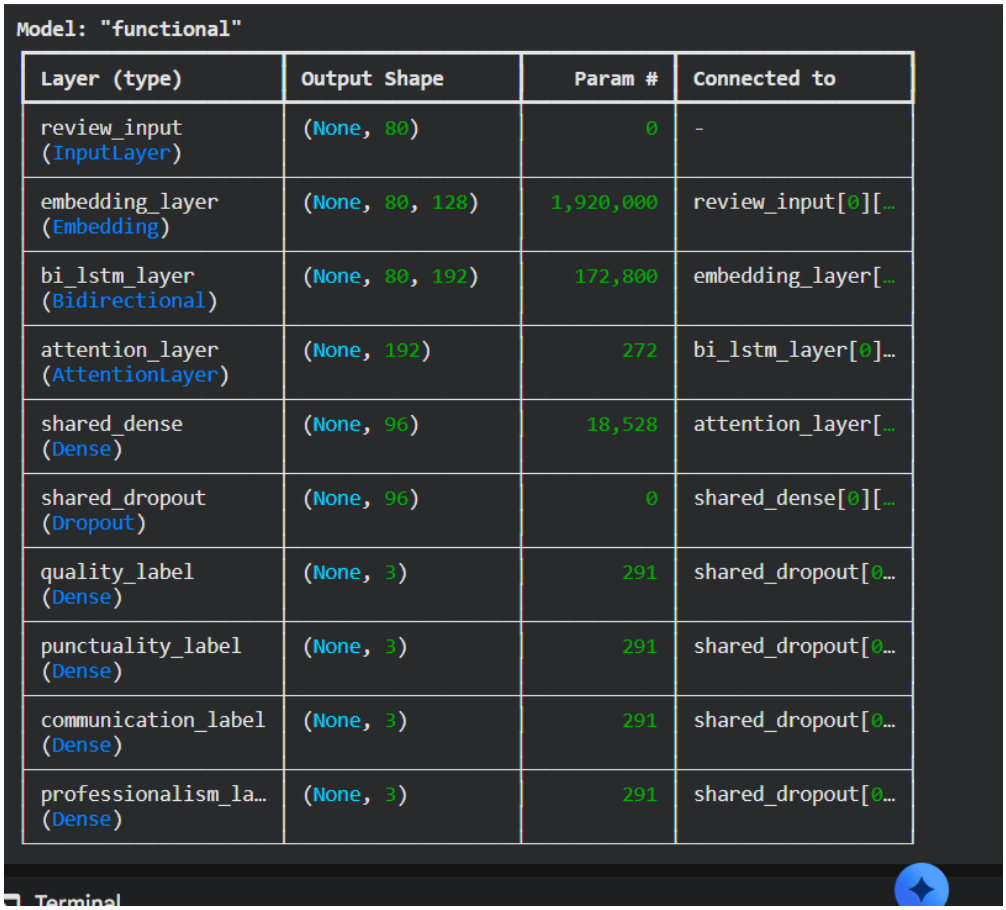
Sentiment Class – One of the three sentiment categories used in this research: Positive, Neutral, or Negative.

11 APPENDICES

11.1 Appendix A Research instruments or extended results

```
Train: (17466, 30)
Validation: (3761, 30)
Test: (3773, 30)
```

Figure 11.1 A.1:Dataset Partition Evidence



Model: "functional"

Layer (type)	Output Shape	Param #	Connected to
review_input (InputLayer)	(None, 80)	0	-
embedding_layer (Embedding)	(None, 80, 128)	1,920,000	review_input[0][…
bi_lstm_layer (Bidirectional)	(None, 80, 192)	172,800	embedding_layer[…
attention_layer (AttentionLayer)	(None, 192)	272	bi_lstm_layer[0]…
shared_dense (Dense)	(None, 96)	18,528	attention_layer[…
shared_dropout (Dropout)	(None, 96)	0	shared_dense[0][…
quality_label (Dense)	(None, 3)	291	shared_dropout[0…
punctuality_label (Dense)	(None, 3)	291	shared_dropout[0…
communication_label (Dense)	(None, 3)	291	shared_dropout[0…
professionalism_la… (Dense)	(None, 3)	291	shared_dropout[0…

Terminal

Figure 11.2 A.2:Implemented Model Summary

```
================================================================================
Test Fake Review Detection Report
================================================================================
              precision    recall  f1-score   support

        Real       0.98      0.97      0.98      3387
        Fake       0.77      0.80      0.79       386

    accuracy                           0.96      3773
   macro avg       0.87      0.89      0.88      3773
weighted avg       0.96      0.96      0.96      3773
```

Figure 11.3 A.3:Review Credibility Evaluation Output

```
Reviews: (25000, 11)
Aspect labels: (25000, 6)
Credibility: (25000, 13)
Provider summary: (4998, 13)
```

	review_id	booking_id	provider_id	user_id	service_type	district	rating	review_text	review_date	verified_booking	dataset_split
0	R00001	B234652	P010234	U34114	Masons	Matara	1	EXCELLENT EXCELLENT EXCELLENT service!!! Best ...	2025-09-21	0	train
1	R00002	B380896	P020134	U71066	Carpenters	Nuwara Eliya	5	Perfect work 100% recommended visit http://ser...	2025-11-27	0	test
2	R00003	B192559	P030016	U12777	Tile Workers	Jaffna	4	For the bathroom tile work, bathroom tile work...	2025-07-26	1	train
3	R00004	B585697	P040029	U72068	Plumbers	Matara	2	During the tap repair job, work quality was po...	2025-07-07	0	train
4	R00005	B246515	P050254	U18115	Electricians	Batticaloa	5	During the switch board job, quality eka honda...	2025-03-19	0	train

	review_id	quality_label	punctuality_label	communication_label	professionalism_label	overall_sentiment_label
0	R00001	Positive	Positive	Neutral	Negative	Neutral
1	R00002	Negative	Positive	Positive	Negative	Neutral
2	R00003	Positive	Positive	Positive	Positive	Positive
3	R00004	Negative	Negative	Negative	Negative	Negative

Figure 11.4 A.4:Dataset Structure Sample

11.2 Appendix B Academic integrity and AI use disclosure

Tool Used: ChatGPT (OpenAI)

Purpose:

ChatGPT was used to support academic language refinement, structural organization, grammar improvement, and the presentation of selected technical explanations within the thesis.

Representative Use:

The tool was used to assist in refining explanations of the implemented methodology, component implementation, experimental results, discussion, and conclusion based on information and evidence provided by the author.

Verification:

AI-assisted content was reviewed and verified against the actual implementation, dataset information, model configurations, experimental outputs, and evaluation results before inclusion in the thesis.

Sections/Artefacts Affected:

AI assistance was primarily used for language refinement, organization, and presentation of selected thesis sections. The implementation, model training, experimental evaluation, technical decisions, and project-specific results were produced from the author's own research and development work.